软件分析

南京大学
计算机科学与技术系
程序设计语言与静态分析研究组
李樾 谭添

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Static Program Analysis

Interprocedural Analysis

Nanjing University

Tian Tan

2021

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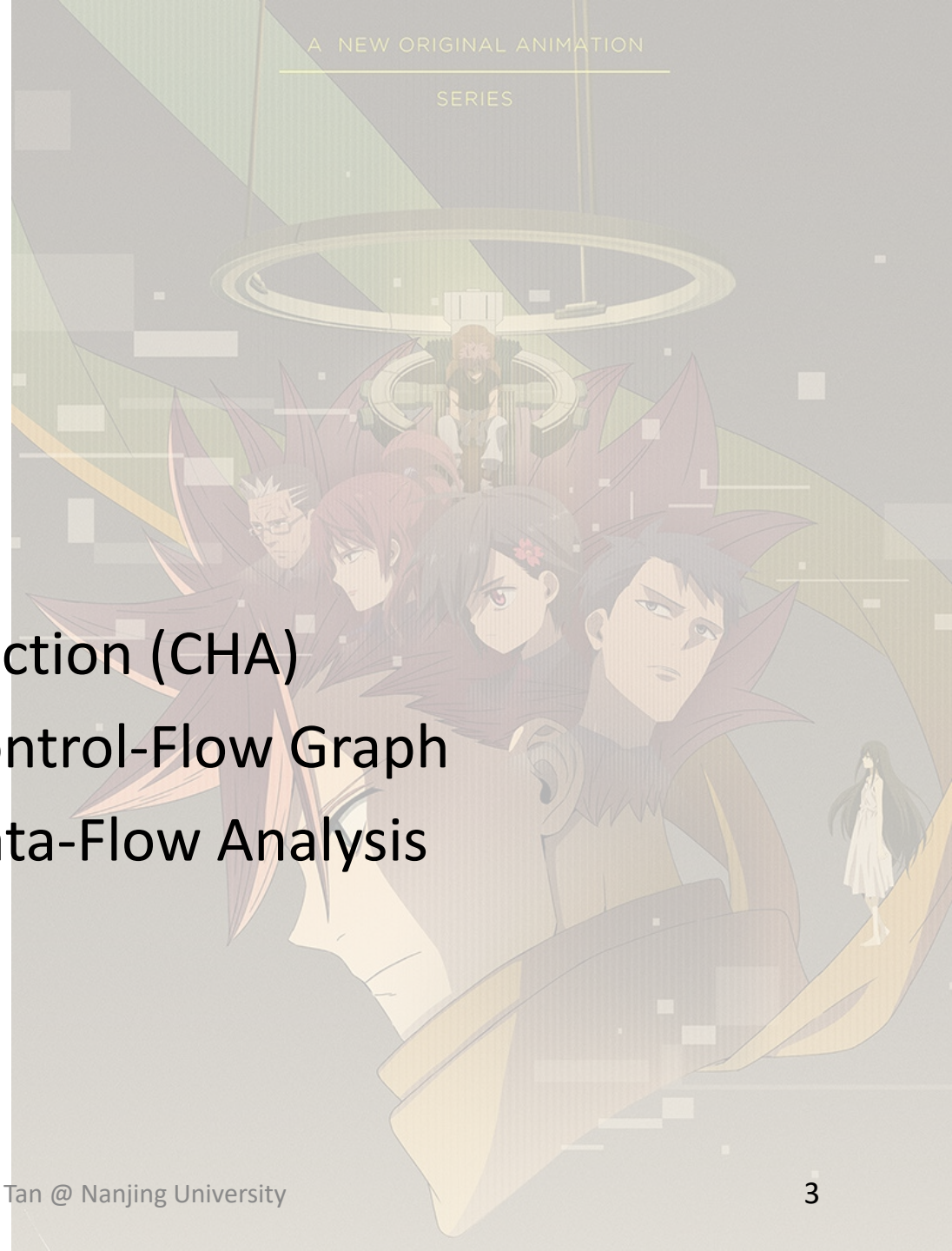
Contents



1. Motivation
2. Call Graph Construction (CHA)
3. Interprocedural Control-Flow Graph
4. Interprocedural Data-Flow Analysis

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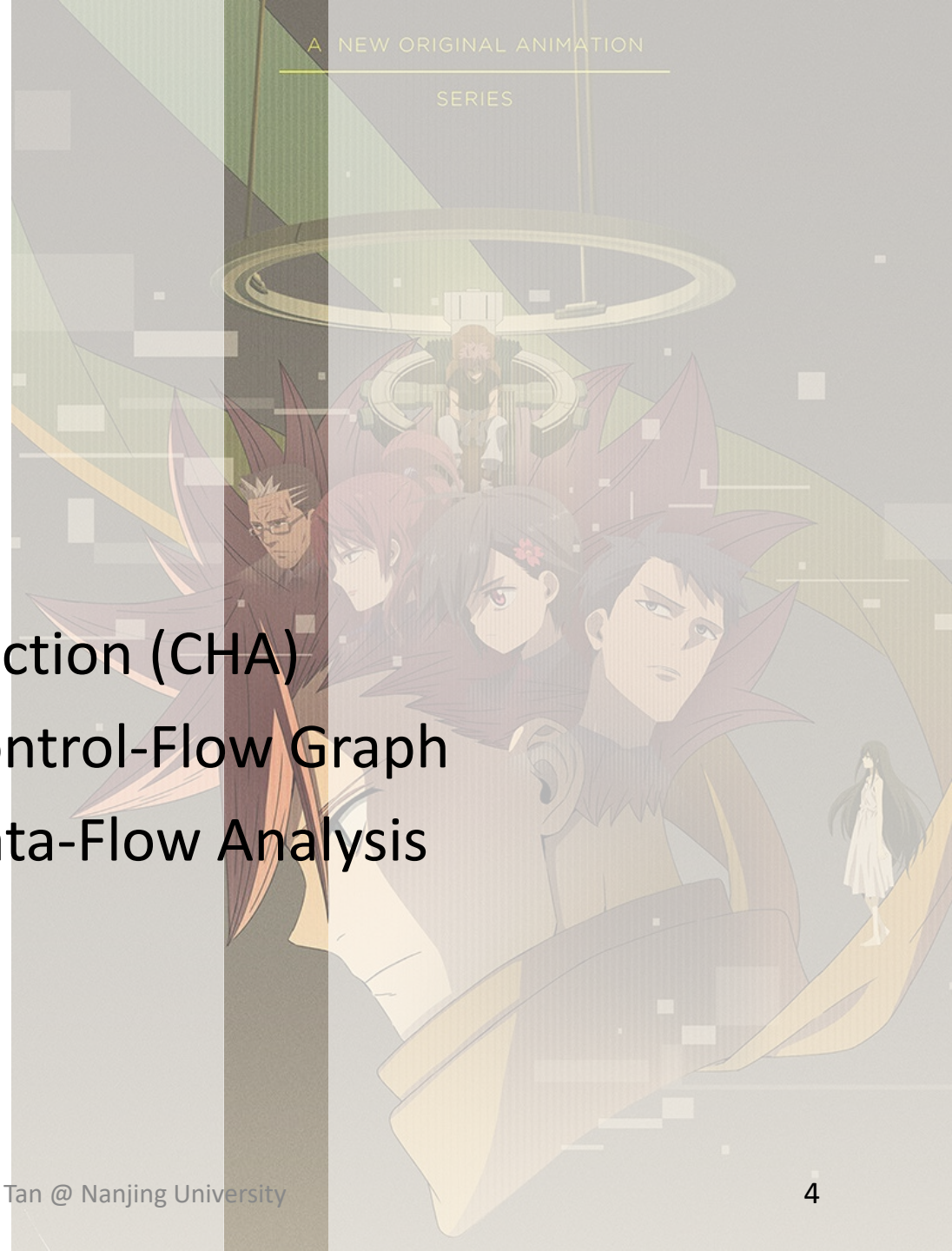

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Motivation of Interprocedural Analysis

Constant Propagation

过程内

So far, all analyses we learnt are intraprocedural.
How to deal with method calls?

```
void foo() {
    int n = bar(42);
}

int bar(int x) {
    int y = x + 1;
    return 10;
}
```

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- Source of imprecision

➢ x = NAC, y = NAC
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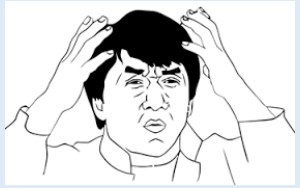

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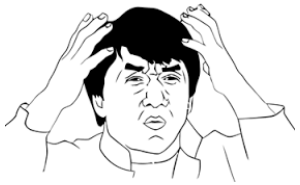

For better precision, we need Interprocedural analysis: propagate data-flow information along **interprocedural control-flow edges** (i.e., call and return edges)

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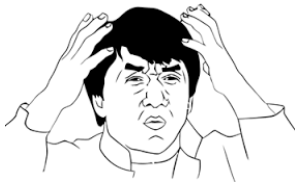

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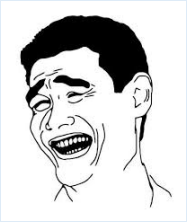

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Constant Propagation

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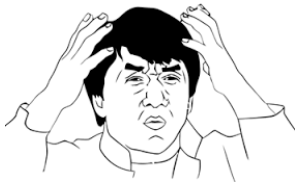

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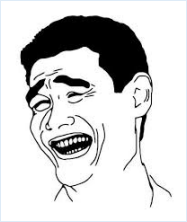
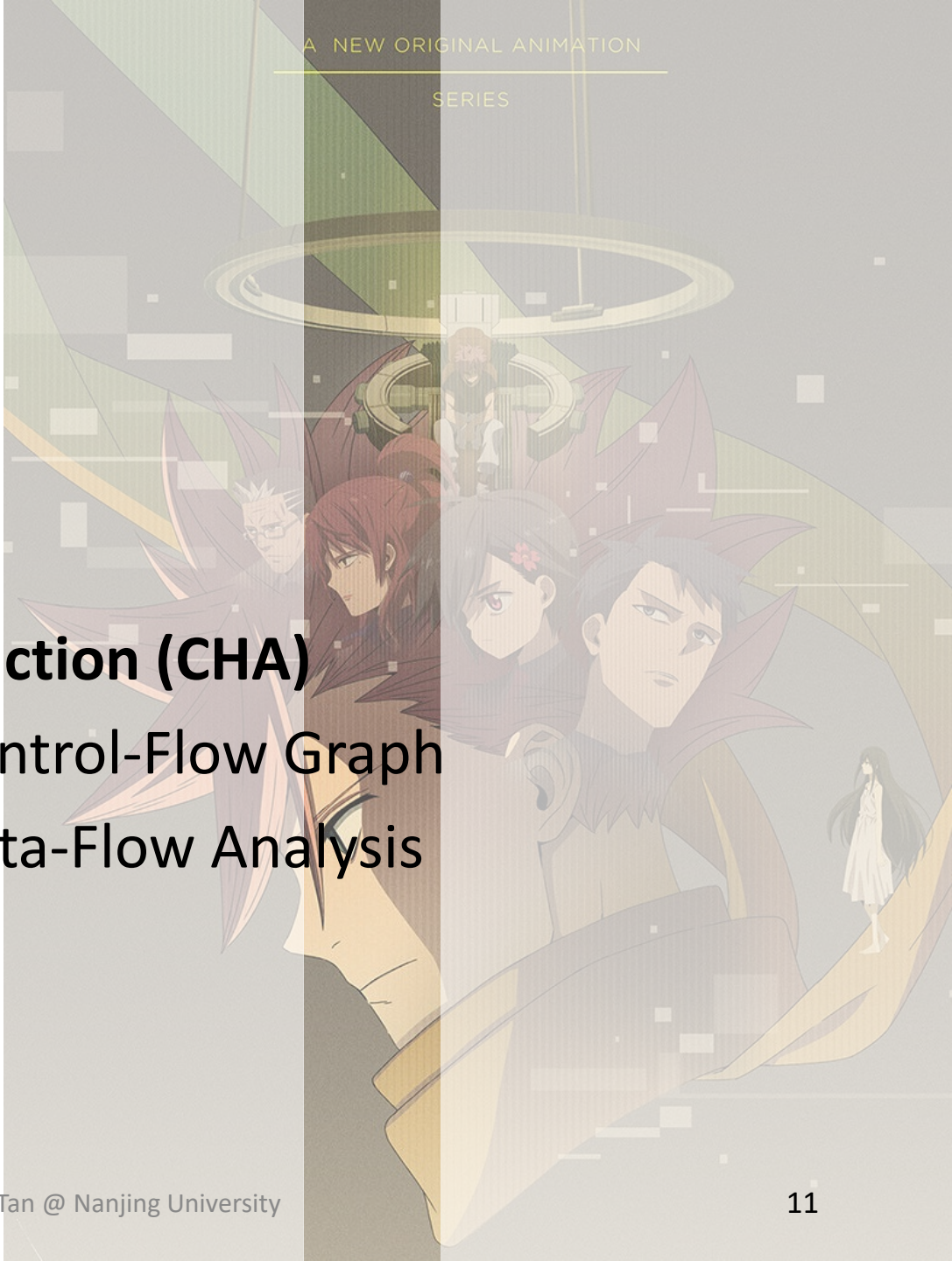


To perform interprocedural analysis, we need **call graph**

Contents



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Call Graph

A representation of calling relationships in the program

- Essentially, a call graph is a set of call edges from call-sites to their target methods (callees)

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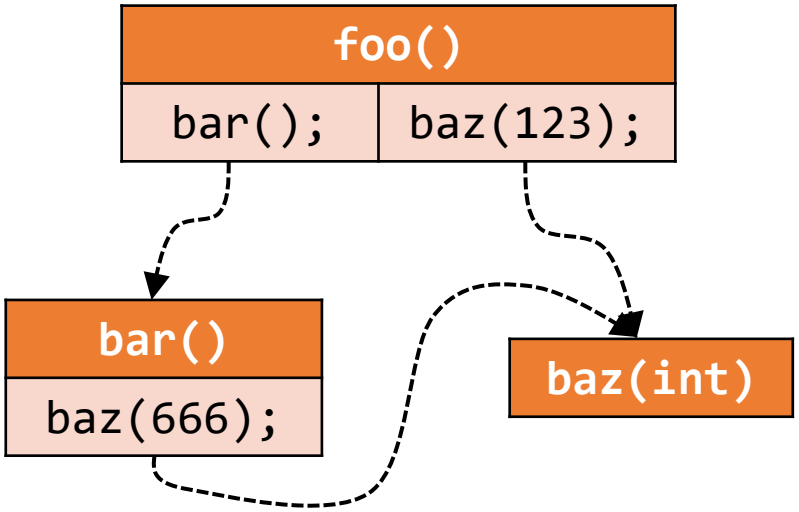
- Essentially, a call graph is a set of call edges from call-sites to their target methods (callees)

```
void foo() {
  bar();
  baz(123);
}

void bar() {
  baz(666);
}

void baz(int x) { }
```



Applications of Call Graph

- Foundation of all interprocedural analyses
- Program optimization
- Program understanding
- Program debugging
- Program testing
- And many more …

Call graph is **VERY important** program information

Call Graph Construction for OOPLs (focus on Java)

本课程主要 针对面向对象语言

针对java

- Class hierarchy analysis (CHA)
- Rapid type analysis (RTA)
- Variable type analysis (VTA)
- Pointer analysis (*k*-CFA)

Call Graph Construction for OOPLs (focus on Java)

More efficient

- Class hierarchy analysis (CHA)
- Rapid type analysis (RTA)
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- Pointer analysis (*k*-CFA)

More precise

Call Graph Construction for OOPLs (focus on Java)

this lecture

More efficient

- **Class hierarchy analysis (CHA)**
- Rapid type analysis (RTA)
- Variable type analysis (VTA)
- **Pointer analysis (*k*-CFA)**

More precise

next lectures

Method Calls (Invocations) in Java

	Static call	Special call	Virtual call
Instruction	`invokestatic`	`invokespecial`	`invokeinterface` `invokevirtual`

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#Target methods	1	1	≥1 (**polymorphism**)
Determinacy	Compile-time	Compile-time	Run-time

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Key to call graph construction for OOPLs

Method Dispatch of Virtual Calls

During run-time, a virtual call is resolved based on

1. type of the receiver object (pointed by o)
2. method signature at the call site

```
o¹.foo(…)²;
```

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In this lecture, a signature acts as an identifier of a method

- Signature = class type + method name + descriptor
- Descriptor = return type + parameter types

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```
class C {
  T foo(P p, Q q, R r) { … }
}
```

```
<C: T foo(P,Q,R)>
```

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During run-time, a virtual call is resolved based on

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class C {
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```

```
<C: T foo(P,Q,R)>
```

`C.foo(P,Q,R)` for short

- Signature = class type + method name + descriptor
- Descriptor = return type + parameter types

Method Dispatch of Virtual Calls

During run-time, a virtual call is resolved based on

1. type of the receiver object (pointed by o): $\boldsymbol{c}$
2. method signature at the call site: $\boldsymbol{m}$

```
o¹.foo(…)²;
```

We define function $\text{Dispatch}(c, m)$ to simulate the procedure of run-time method dispatch

在自己类中找

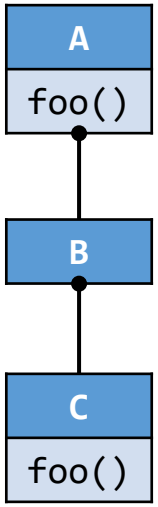
$$\text{Dispatch}(c, m) = \begin{cases} m', & \textbf{if } c \text{ contains non-abstract method } m' \text{ that has the same } \underline{\text{name}} \text{ and } \underline{\text{descriptor}} \text{ as } m \\ \text{Dispatch}(c', m), & \textbf{otherwise} \end{cases}$$

where c' is superclass of c

递归往上找

```
<C: T foo(P,Q,R)>
```

Dispatch: An Example




```
class A {
  void foo() {…}
}

class B extends A {
}

class C extends B {
  void foo() {…}
}

void dispatch() {
  A x = new B();
  x.foo();?

  A y = new C();
  y.foo();
}
```

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Dispatch(B, A.foo()) = ?

<B : void foo()>

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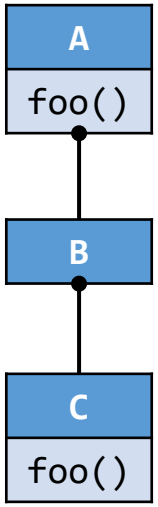


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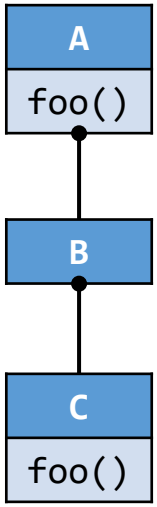
Dispatch(B, `A.foo()`) = `A.foo()`

Dispatch(C, `A.foo()`) = ?

Dispatch: An Example

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Dispatch(B, `A.foo()`) = `A.foo()`

Dispatch(C, `A.foo()`) = `C.foo()`

Class Hierarchy Analysis* (CHA)

- Require the class hierarchy information (inheritance structure) of the whole program
- Resolve a virtual call based on the declared type of receiver variable of the call site

```
A a = …
a.foo();
```

a可能指向的是 A以及A所有的子类

*Jeffrey Dean, David Grove, Craig Chambers, "*Optimization of Object-Oriented Programs Using Static Class Hierarchy Analysis*". ECOOP 1995.

Class Hierarchy Analysis* (CHA)

- Require the class hierarchy information (inheritance structure) of the whole program
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- Assume the receiver variable a may point to objects of class A or all subclasses of A
  - Resolve target methods by looking up the class hierarchy of class A

mapreduce

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Call Resolution of CHA

We define function Resolve(cs) to resolve possible target methods of a call site cs by CHA

```
Resolve(cs)
    T = {}
    m = method signature at cs
    if cs is a static call then
        T = { m }
    if cs is special call then
        c^m = class type of m
        T = { Dispatch(c^m, m) }
    if cs is a virtual call then
        c = declared type of receiver variable at cs
        foreach c' that is a subclass of c or c itself do
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```
class C {
  static T foo(P p, Q q)
  {…}
}

C.foo(x, y);
```

cs `C.foo(x, y);`

m `<C: T foo(P,Q)>`

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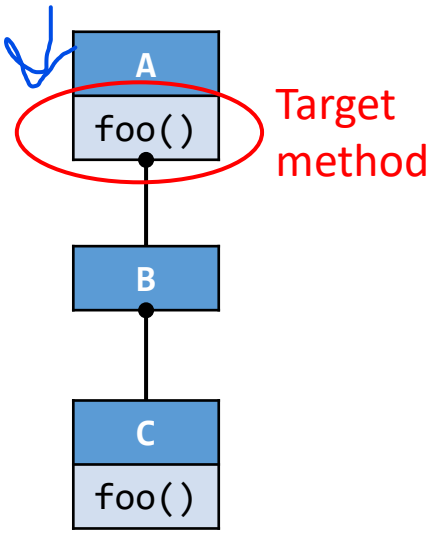
```
class C extends B {
  T foo(P p, Q q) {
    …
    super.foo(p, q);
  }
}
```

为什么还用 dispatch呢?

cs	`super.foo(p, q);`
m	`<B: T foo(P,Q)>`
c^m	B

Call Resolution of CHA



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```

cs	`super.foo(p, q);`
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c^m	`B`

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    c = declared type of receiver variable at cs
    foreach c' that is a subclass of c or c itself **do**
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```
class C extends B {
  T foo(P p, Q q) {
    …
    this.bar();
  }
  private T bar()
}
C c = new C();
```

Special call
- Private instance method
- Constructor
- Superclass instance method

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    foreach c' that is a subclass of c or c itself **do**
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  return T

```
class A {
  T foo(P p, Q q) {…}
}
A a = …
a.foo(x, y);
```

cs	`a.foo(x, y);`
m	`<A: T foo(P,Q)>`
c	`A`

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}
A a = …
a.foo(x, y);
```

cs	`a.foo(x, y);`
m	`<A: T foo(P,Q)>`
c	`A`

Subclasses includes all **direct** and **indirect** subclasses of c

CHA: An Example

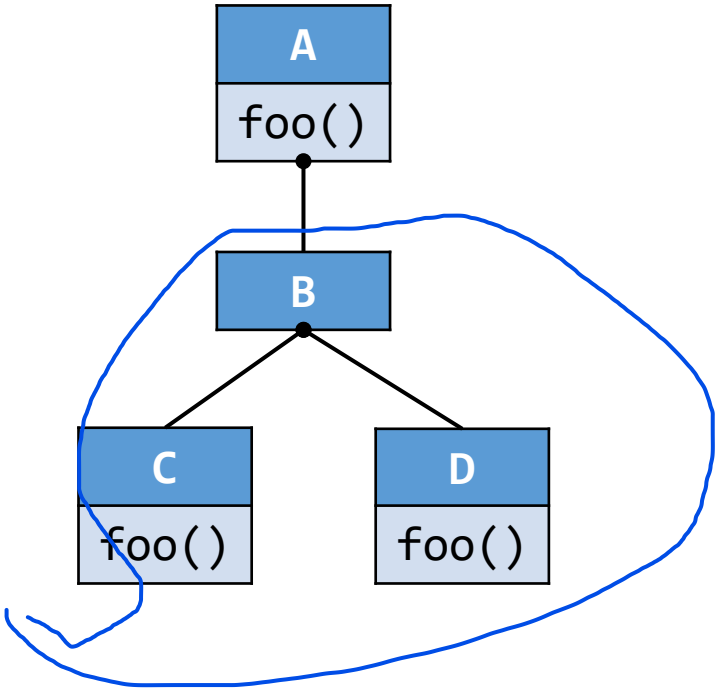
```
class A {
  void foo() {…}
}
class B extends A {}

class C extends B {
  void foo() {…}
}
class D extends B {
  void foo() {…}
}

void resolve() {
  C c = …
  c.foo();

  A a = …
  a.foo();

  B b = …
  b.foo();
}
```



CHA: An Example

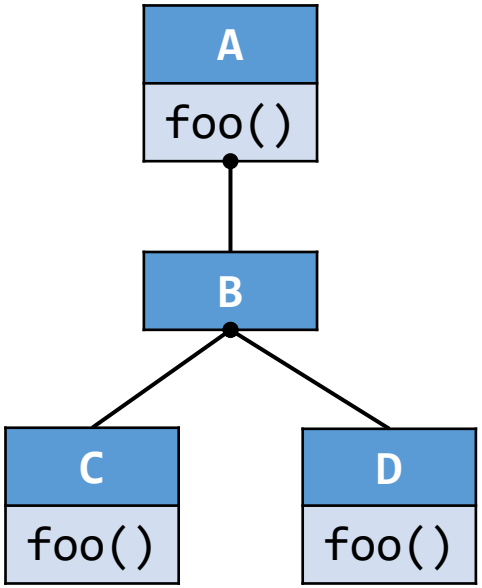
```
class A {
  void foo() {…}
}
class B extends A {}

class C extends B {
  void foo() {…}
}
class D extends B {
  void foo() {…}
}

void resolve() {
  C c = …
  c.foo();?

  A a = …
  a.foo();

  B b = …
  b.foo();
}
```



Resolve(`c.foo()`) = ?

CHA: An Example

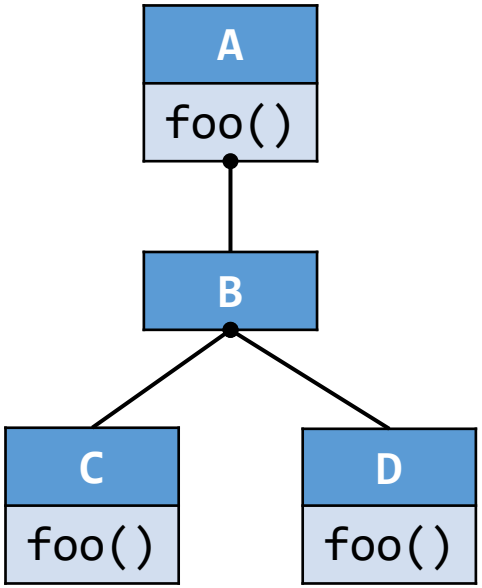
```
class A {
  void foo() {…}
}
class B extends A {}

class C extends B {
  void foo() {…}
}
class D extends B {
  void foo() {…}
}

void resolve() {
  C c = …
  c.foo();

  A a = …
  a.foo();?

  B b = …
  b.foo();
}
```



Resolve(c.foo()) = {C.foo()}

Resolve(a.foo()) = ?

CHA: An Example

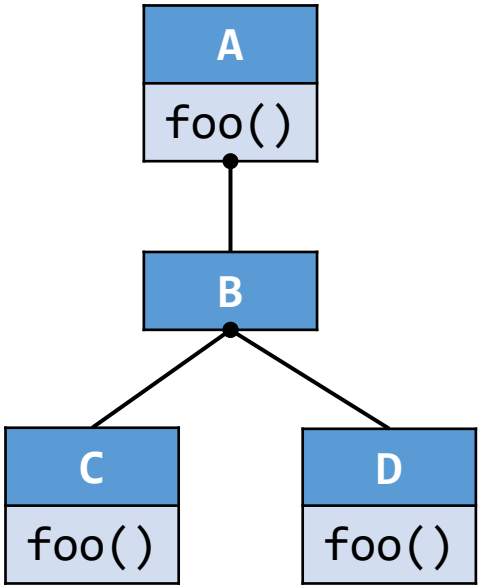
```
class A {
  void foo() {…}
}
class B extends A {}

class C extends B {
  void foo() {…}
}
class D extends B {
  void foo() {…}
}

void resolve() {
  C c = …
  c.foo();

  A a = …
  a.foo();

  B b = …
  b.foo(); ?
}
```



Resolve(c.foo()) = {C.foo()}

Resolve(a.foo()) = {A.foo(), C.foo(), D.foo()}

Resolve(b.foo()) = ?

CHA: An Example

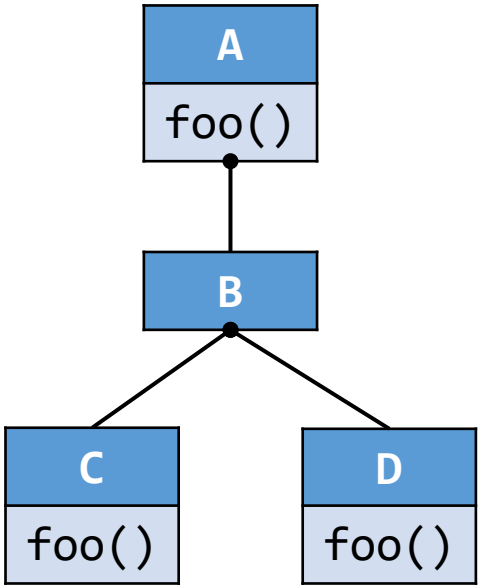
```
class A {
  void foo() {…}
}
class B extends A {}

class C extends B {
  void foo() {…}
}
class D extends B {
  void foo() {…}
}

void resolve() {
  C c = …
  c.foo();

  A a = …
  a.foo();

  B b = …
  b.foo();
}
```



Resolve(c.foo()) = {C.foo()}

Resolve(a.foo()) = {A.foo(), C.foo(), D.foo()}

Resolve(b.foo()) = {A.foo(), C.foo(), D.foo()}

CHA: An Example

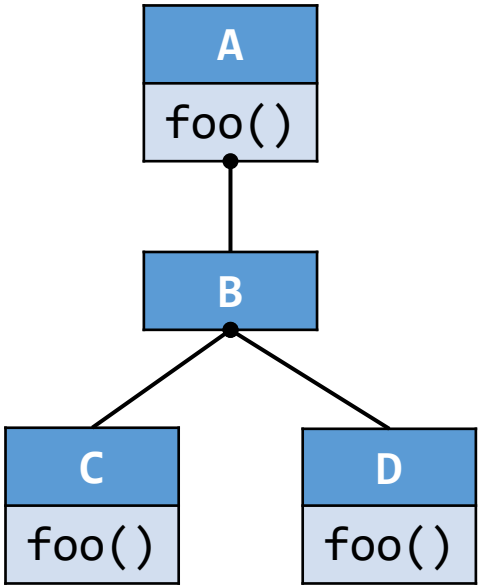
```
class A {
  void foo() {…}
}
class B extends A {}

class C extends B {
  void foo() {…}
}
class D extends B {
  void foo() {…}
}

void resolve() {
  C c = …
  c.foo();

  A a = …
  a.foo();

  B b = new B();
  b.foo();
}
```



Resolve(c.foo()) = {C.foo()}

Resolve(a.foo()) = {A.foo(), C.foo(), D.foo()}

Resolve(b.foo()) = ?

CHA: An Example

.foo()是不是一个virtual call呢?

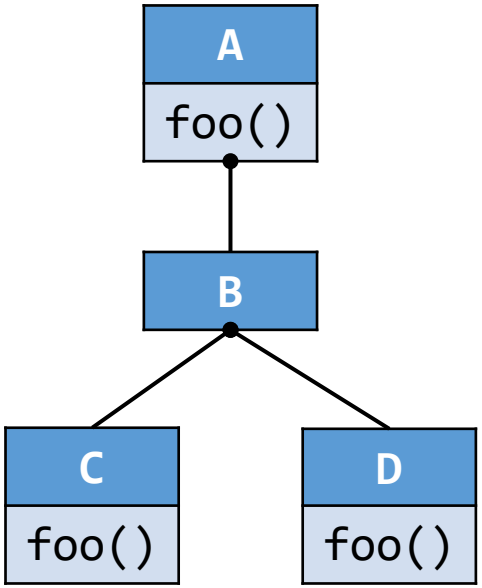
```
class A {
  void foo() {…}
}
class B extends A {}

class C extends B {
  void foo() {…}
}
class D extends B {
  void foo() {…}
}

void resolve() {
  C c = …
  c.foo();

  A a = …
  a.foo();

  B b = new B();
  b.foo();
}
```



Resolve(c.foo()) = {C.foo()}

Resolve(a.foo()) = {A.foo(), C.foo(), D.foo()}

因为CHA 仅仅根据 声明类型做的

错的、假的

Resolve(b.foo()) = {A.foo(), C.foo(), D.foo()}

Spurious call targets

Features of CHA

- Advantage: fast
  - Only consider the declared type of receiver variable at the call-site, and its inheritance hierarchy
  - Ignore data- and control-flow information

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  - Addressed in next lectures

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  - Addressed in next lectures

Common usage: IDE

CHA in IDE (IntelliJ IDEA)

idea 用的就是 CHA的方式！！！！

TestCHA.java

```
1  public class TestCHA {
2      void test() {
3          B b = new B();
4          b.foo();
5      }
6  }
7  class A {
8      void foo() {}
9  }
10 class B extends A {}
11 class C extends B {
12     void foo() {}
13 }
14 class D extends B {
15     void foo() {}
16 }
```

Hierarchy: Callees of foo

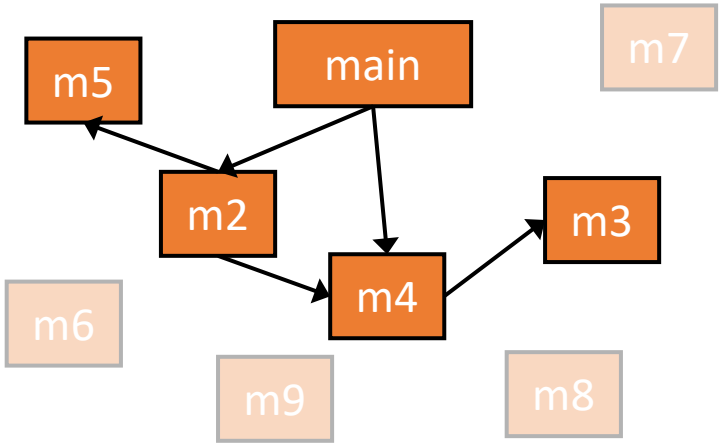
Scope: All

- A.foo() ()
  - C.foo() ()
  - D.foo() ()

Call Graph Construction

Build call graph for whole program via CHA

- Start from entry methods (focus on main method)
- For each reachable method m, resolve target methods for each call site cs in m via CHA ($\text{Resolve}(cs)$)
- Repeat until no new method is discovered



Call Graph Construction: Algorithm

```
BuildCallGraph(m^entry)
  WL = [m^entry], CG = {}, RM = {}
  while WL is not empty do
    remove m from WL
    if m ∉ RM then
      add m to RM
      foreach call site cs in m do
        T = Resolve(cs)
        foreach target method m' in T do
          add cs → m' to CG
          add m' to WL
  return CG
```

WL	Work list, containing the methods to be processed
CG	Call graph, a set of call edges
RM	A set of reachable methods

Call Graph Construction: Algorithm

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Initialize the algorithm

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          add cs → m' to CG
          add m' to WL
  return CG
```

Initialize the algorithm

Resolve target methods via CHA

WL	Work list, containing the methods to be processed
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Call Graph Construction: Algorithm

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        foreach target method m' in T do
          add cs → m' to CG
          add m' to WL
  return CG
```

Initialize the algorithm

Resolve target methods via CHA

Add call edges to call graph

WL	Work list, containing the methods to be processed
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Call Graph Construction: Algorithm

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          add cs → m' to CG
          add m' to WL
  return CG
```

Initialize the algorithm

Resolve target methods via CHA

Add call edges to call graph

May discover new method, add it to work list

WL	Work list, containing the methods to be processed
CG	Call graph, a set of call edges
RM	A set of reachable methods

Call Graph Construction: An Example

```
class A {
  static void main() {
    A.foo();
  }
  static void foo() {
    A a = new A();
    a.bar();
  }
  void bar() {
    C c = new C();
    c.bar();
  }
}
class B extends A {
  void bar() {} }
class C extends A {
  void bar() {
    if (…) A.foo();
  }
  void m() {}
}
```

A.main()
A.foo();

A.foo()
a.bar();

A.bar()
c.bar();

B.Bar()

C.bar()
A.foo();

C.m()

Initialization with main method

$WL = [\texttt{A.main()}]$

Call Graph Construction: An Example

```
class A {
  static void main() {
    A.foo();
  }
  static void foo() {
    A a = new A();
    a.bar();
  }
  void bar() {
    C c = new C();
    c.bar();
  }
}
class B extends A {
  void bar() {} }
class C extends A {
  void bar() {
    if (…) A.foo();
  }
  void m() {}
}
```

A.main()
A.foo();

A.foo()
a.bar();

A.bar()
c.bar();

B.Bar()

C.bar()
A.foo();

C.m()

$WL = [\ \ ]$

Resolve(A.foo()) = ?

Call Graph Construction: An Example

```
class A {
  static void main() {
    A.foo();
  }
  static void foo() {
    A a = new A();
    a.bar();
  }
  void bar() {
    C c = new C();
    c.bar();
  }
}
class B extends A {
  void bar() {} }
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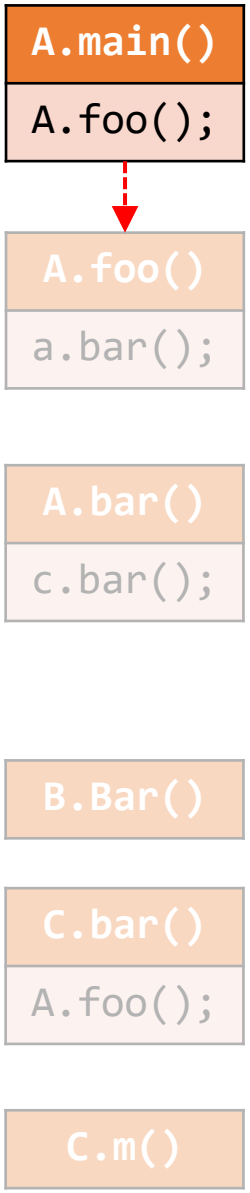
  void bar() {
    if (…) A.foo();
  }
  void m() {}
}
```



$WL = [\texttt{A.foo()}]$

$\text{Resolve}(\texttt{A.foo()}) = \{\ \texttt{A.foo()}\ \}$

Call Graph Construction: An Example

```
class A {
  static void main() {
    A.foo();
  }
  static void foo() {
    A a = new A();
    a.bar();
  }
  void bar() {
    C c = new C();
    c.bar();
  }
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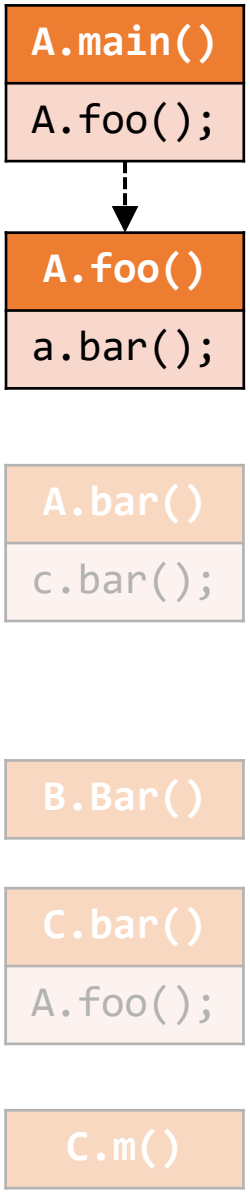
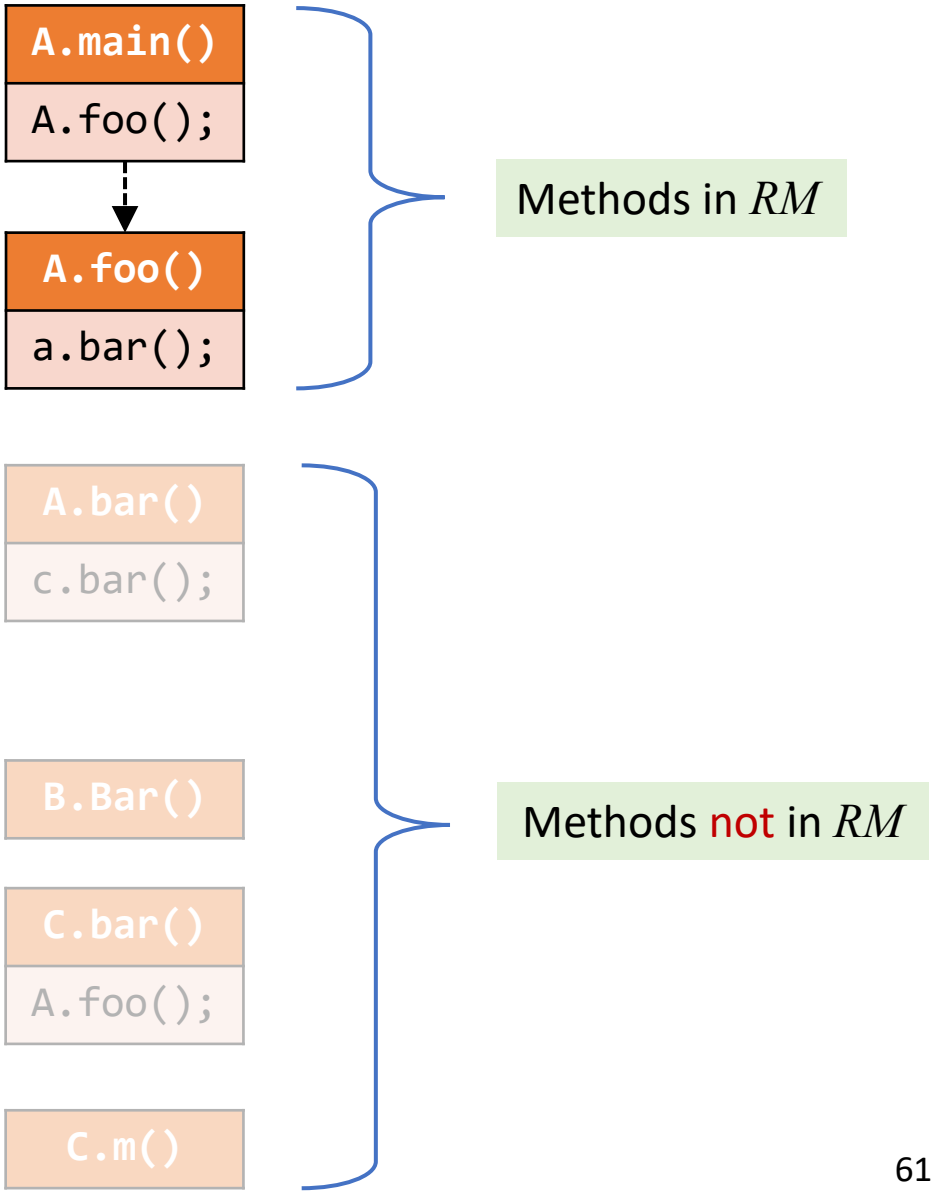


$WL = [\ \ ]$

Call Graph Construction: An Example

```
class A {
  static void main() {
    A.foo();
  }
  static void foo() {
    A a = new A();
    a.bar();
  }
  void bar() {
    C c = new C();
    c.bar();
  }
}
class B extends A {
  void bar() {} }
class C extends A {
  void bar() {
    if (…) A.foo();
  }
  void m() {}
}
```



Call Graph Construction: An Example

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class A {
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    A.foo();
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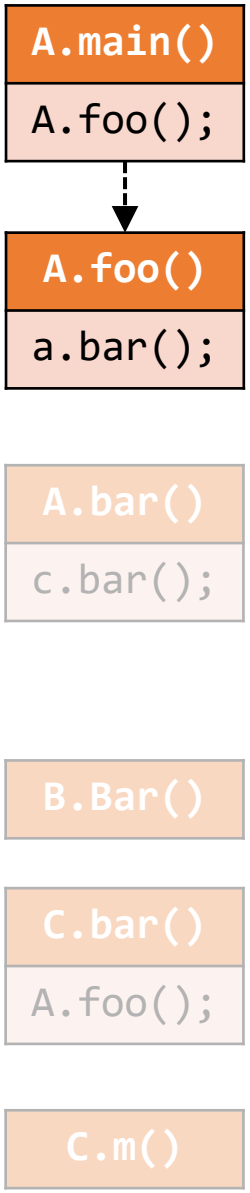
  void bar() {
    if (…) A.foo();
  }
  void m() {}
}
```



$WL = [\ \ ]$

Resolve(a.bar()) = ?

Call Graph Construction: An Example

```
class A {
  static void main() {
    A.foo();
  }
  static void foo() {
    A a = new A();
    a.bar();
  }
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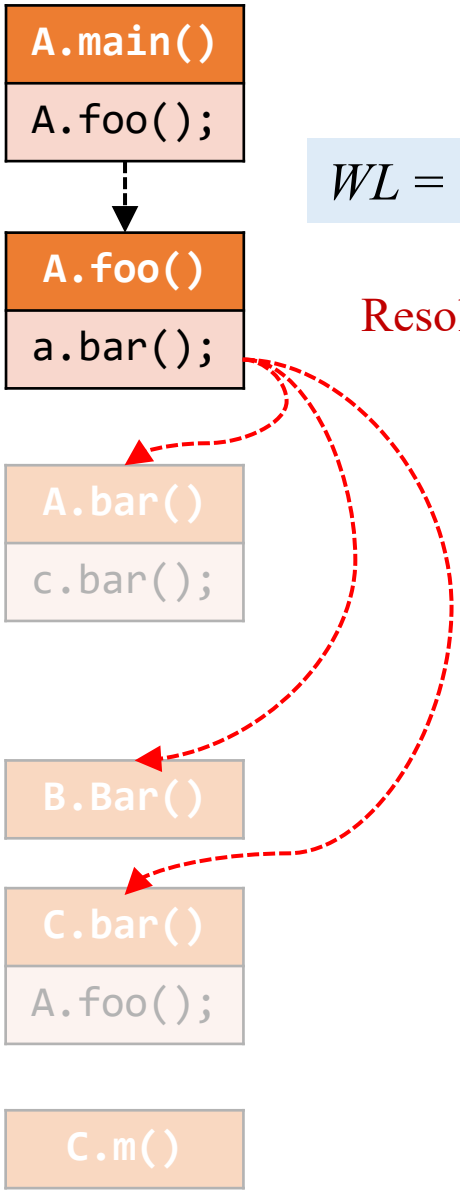

$WL = [$`A.bar()`, `B.bar()`, `C.bar()`$]$

Resolve(`a.bar()`) = { `A.bar()`, `B.bar()`, `C.bar()` }

Call Graph Construction: An Example

```
class A {
  static void main() {
    A.foo();
  }
  static void foo() {
    A a = new A();
    a.bar();
  }
  void bar() {
    C c = new C();
    c.bar();
  }
}
class B extends A {
  void bar() {} }
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  void bar() {
    if (…) A.foo();
  }
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}
```
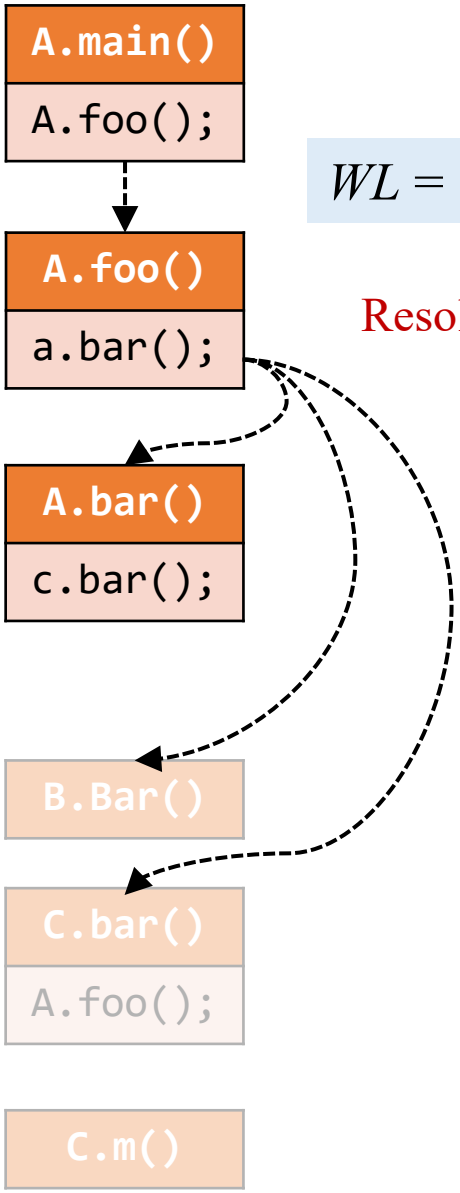
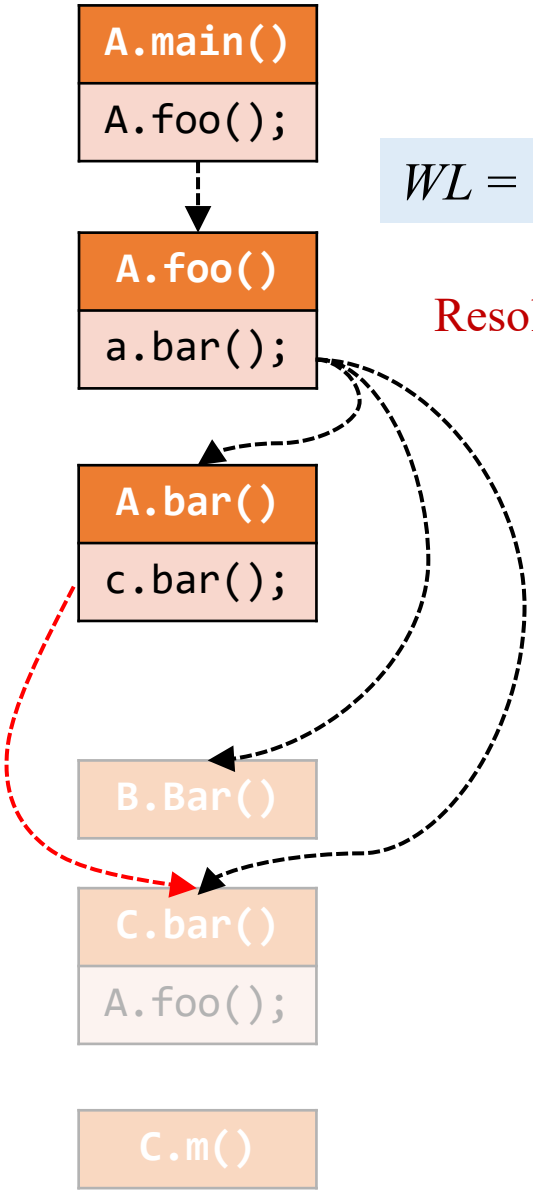



$WL = [\texttt{B.bar()}, \texttt{C.bar()}]$

Resolve(`c.bar()`) = ?

Call Graph Construction: An Example

```
class A {
  static void main() {
    A.foo();
  }
  static void foo() {
    A a = new A();
    a.bar();
  }
  void bar() {
    C c = new C();
    c.bar();
  }
}
class B extends A {
  void bar() {} }
class C extends A {
  void bar() {
    if (…) A.foo();
  }
  void m() {}
}
```



$WL = [\texttt{B.bar()}, \texttt{C.bar()}, \texttt{C.bar()}]$

$\text{Resolve}(\texttt{c.bar()}) = \{\ \texttt{C.bar()}\ \}$

Call Graph Construction: An Example

```
class A {
  static void main() {
    A.foo();
  }
  static void foo() {
    A a = new A();
    a.bar();
  }
  void bar() {
    C c = new C();
    c.bar();
  }
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class C extends A {
  void bar() {
    if (…) A.foo();
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  void m() {}
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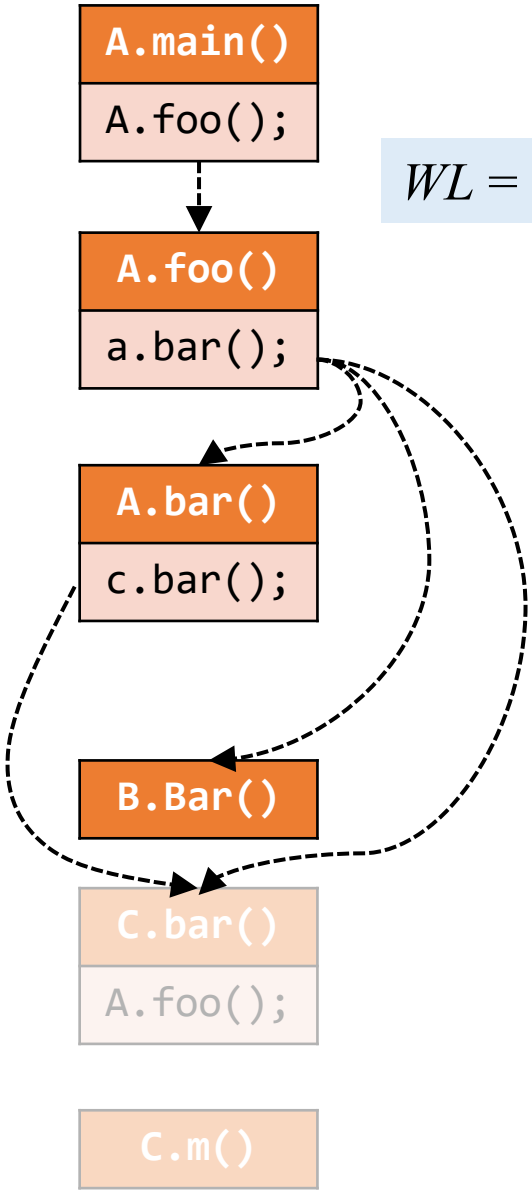

$WL = [\texttt{C.bar()},\texttt{C.bar()}]$

Call Graph Construction: An Example

```
class A {
  static void main() {
    A.foo();
  }
  static void foo() {
    A a = new A();
    a.bar();
  }
  void bar() {
    C c = new C();
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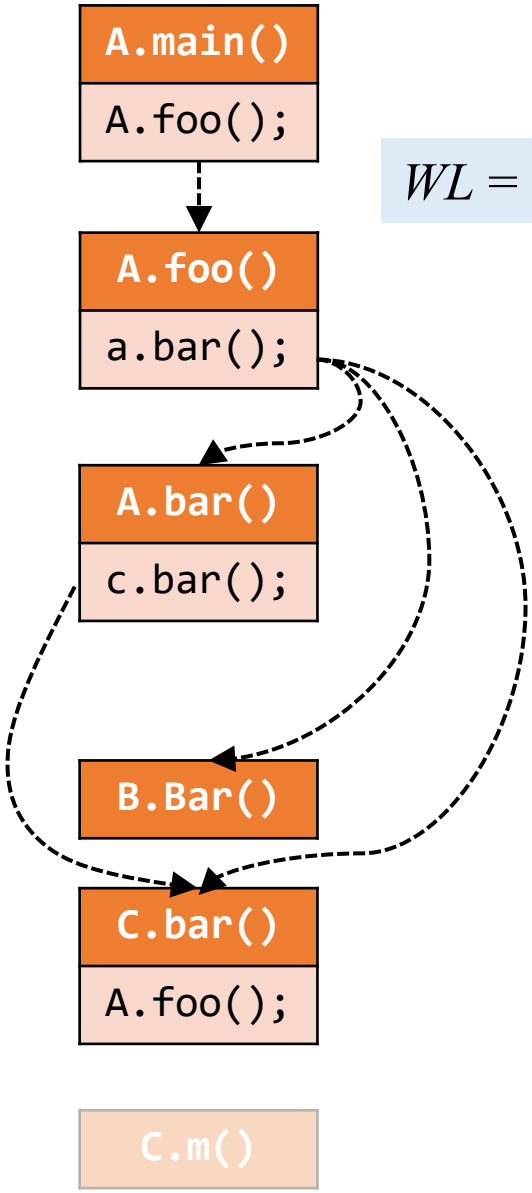

$WL = [\texttt{C.bar()}]$

Call Graph Construction: An Example

```
class A {
  static void main() {
    A.foo();
  }
  static void foo() {
    A a = new A();
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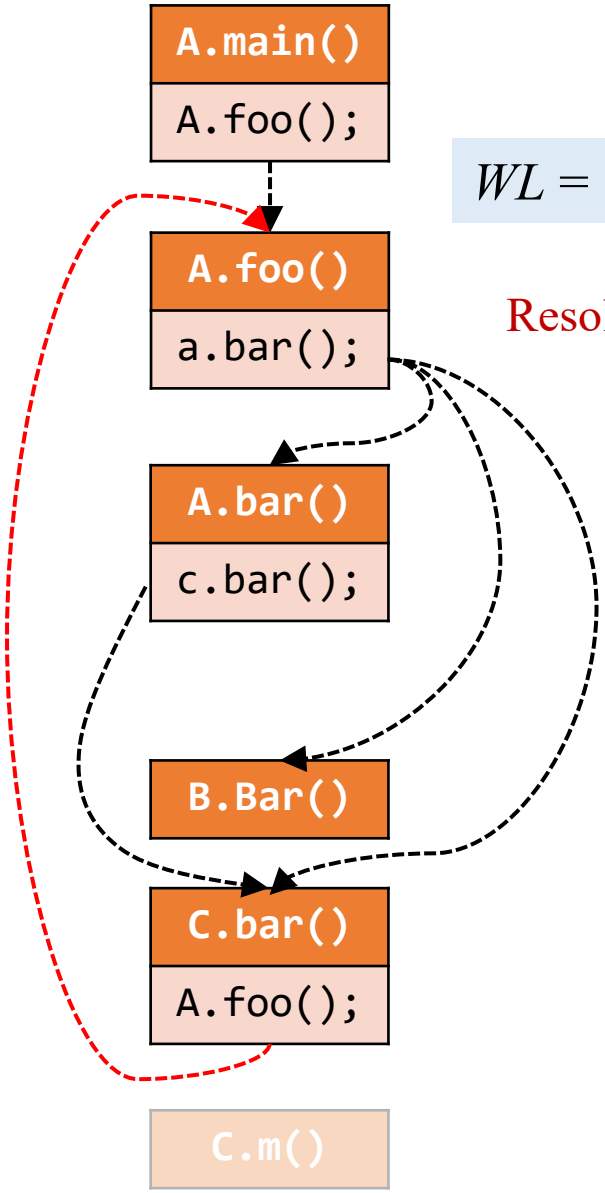

WL = [C.bar(),A.foo()]

Resolve(A.foo()) = { A.foo() }

Call Graph Construction: An Example

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class A {
  static void main() {
    A.foo();
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    A a = new A();
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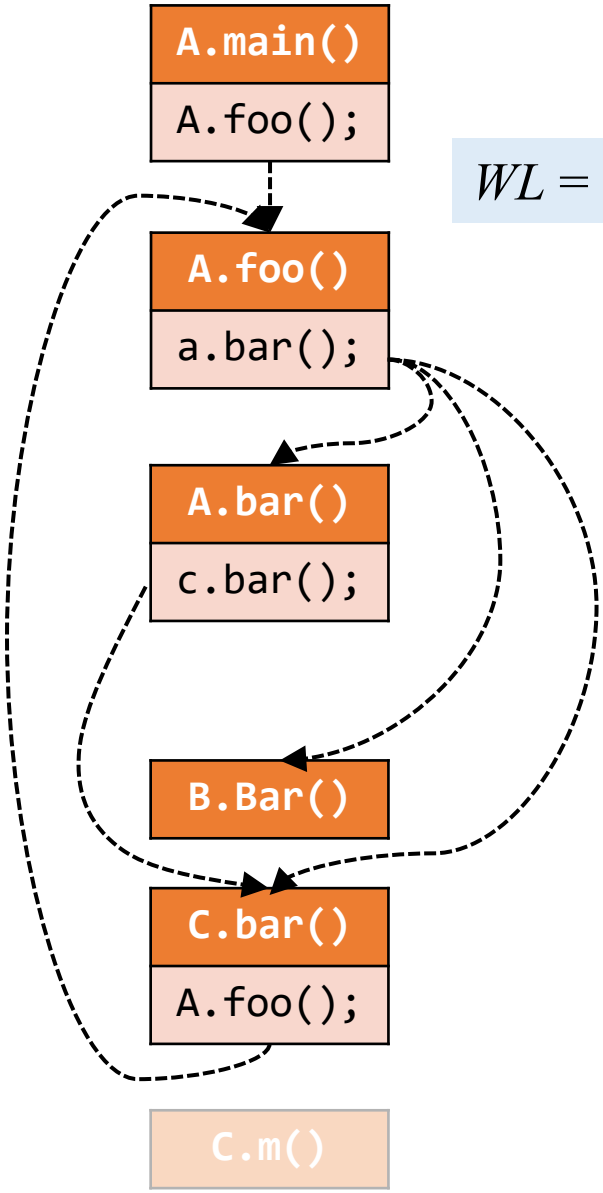

$WL = [\texttt{A.foo()}]$

Call Graph Construction: An Example

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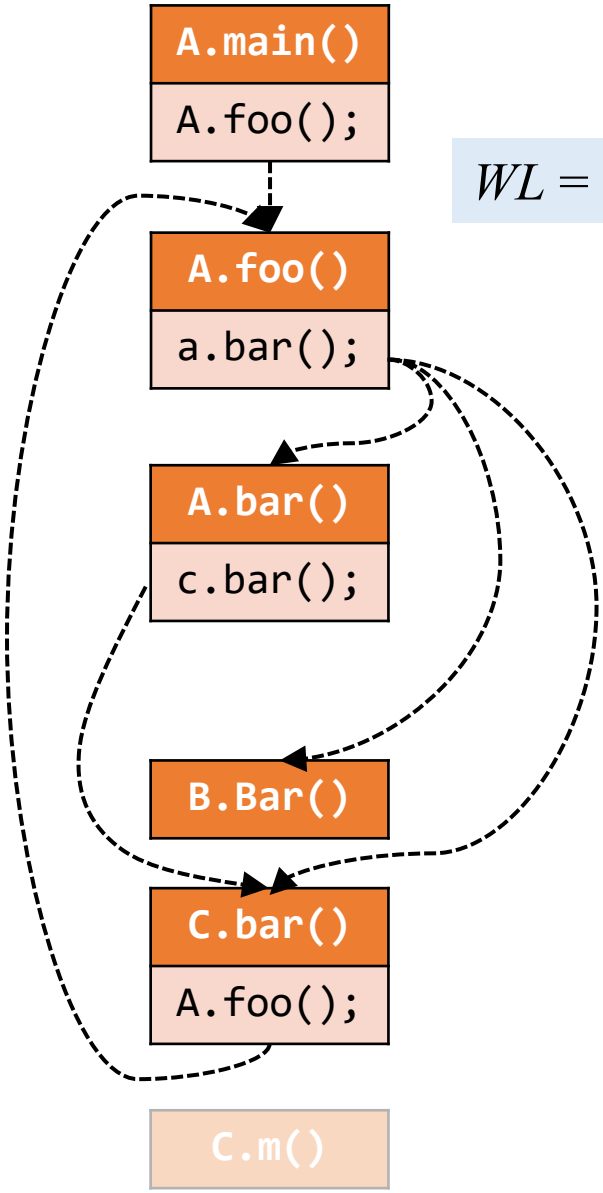

$WL = [\ ]$

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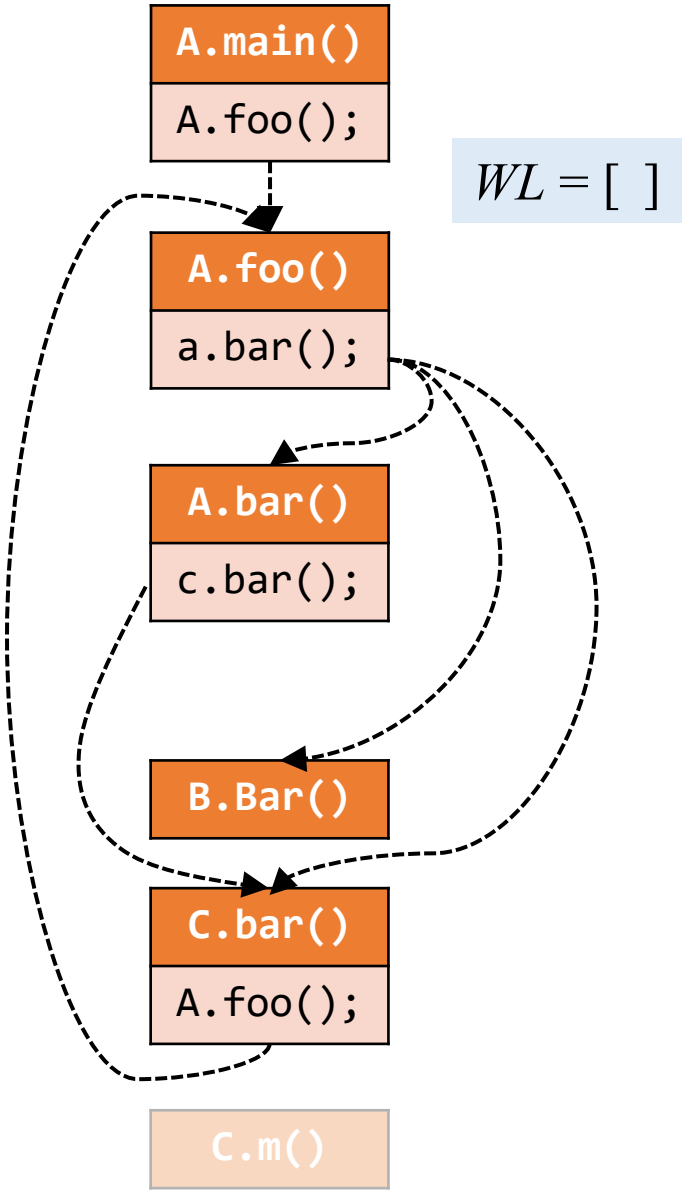

$WL = [\ ]$

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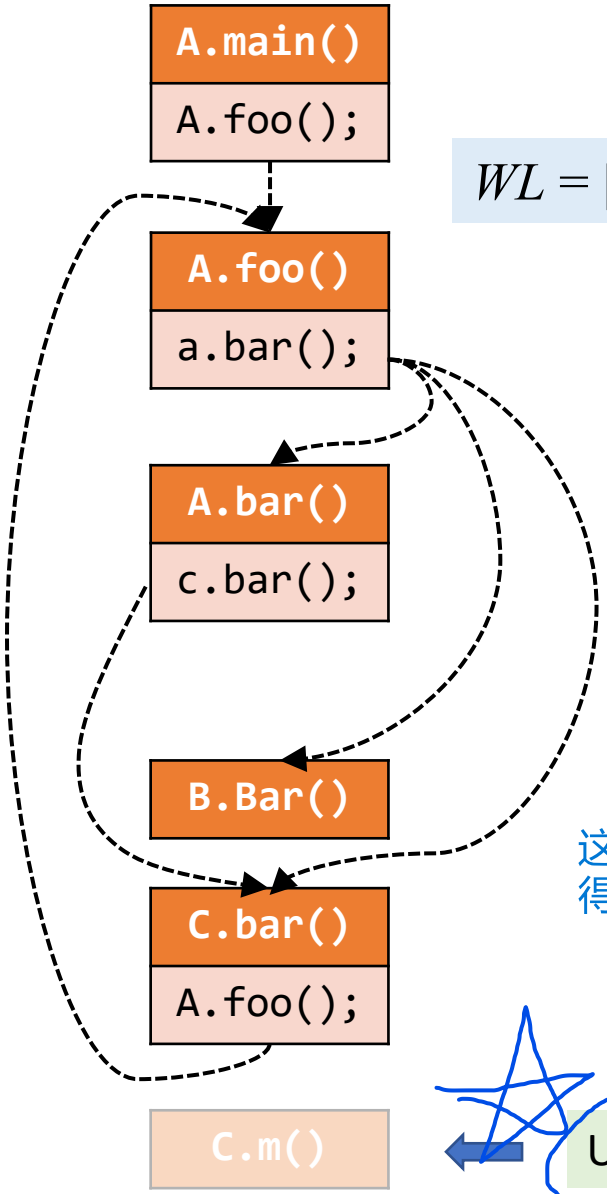


$WL = [\ ]$

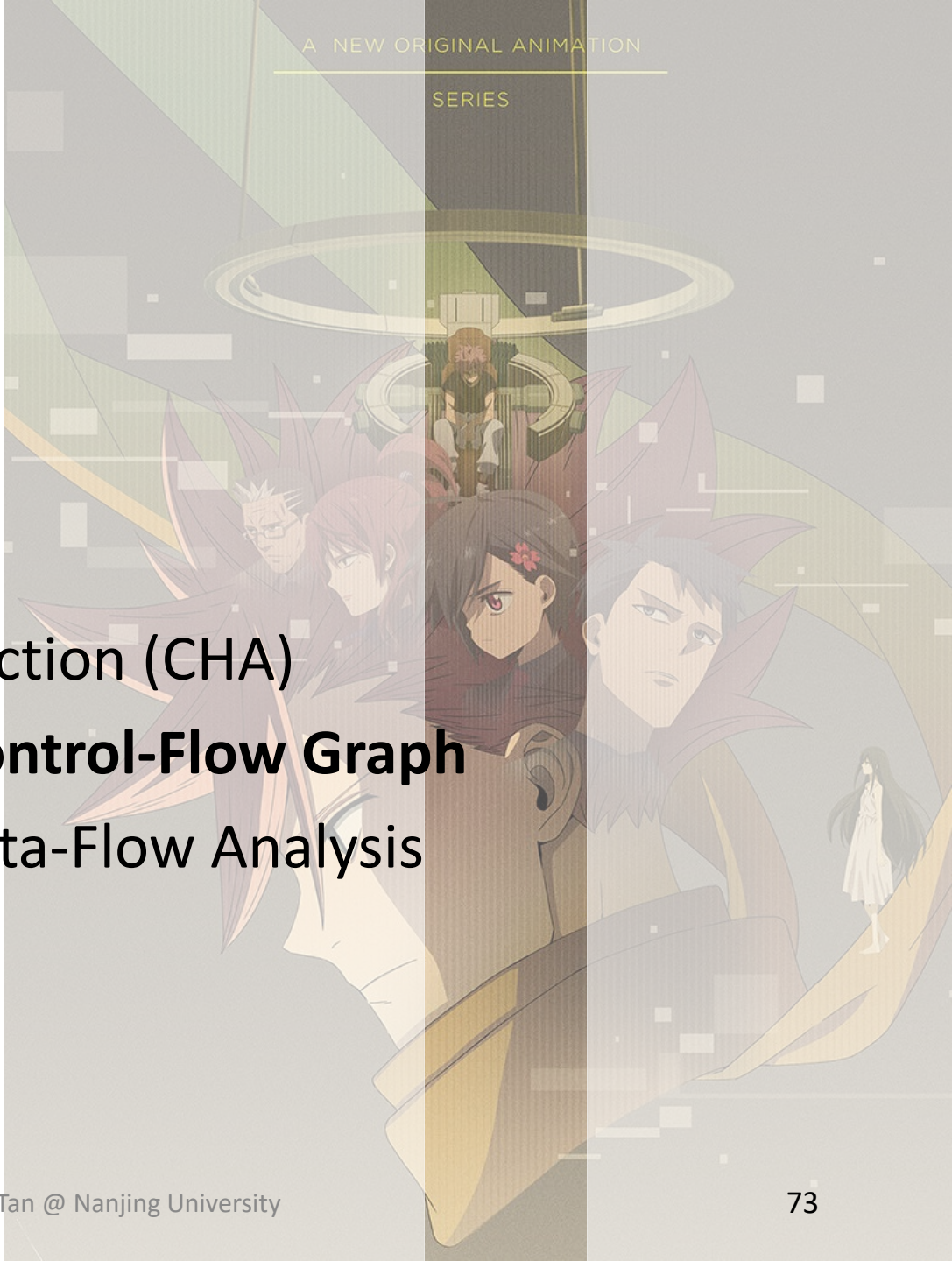
这不是成环了吗?

但是仔细看程序本身，
执行却不会成环

这是通过 CHA构建方法的
得到的 imprecise

Unreachable (dead code)

Contents



1. Motivation
2. Call Graph Construction (CHA)
3. **Interprocedural Control-Flow Graph**
4. Interprocedural Data-Flow Analysis

Interprocedural Control-Flow Graph

- CFG represents structure of an individual method
- ICFG represents structure of the whole program
  - With ICFG, we can perform interprocedural analysis

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```
void foo() {
  bar(…);     // call site
  int n = 3; // return site
}
```

Interprocedural Control-Flow Graph

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ICFG = CFGs + call & return edges

The information for connecting these two kinds of edges comes from call graph

call graph 可以给出这些连接信息

ICFG: An Example

```
static void main() {
  int a, b, c;
  a = 6;
  b = addOne(a);
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  b = ten();
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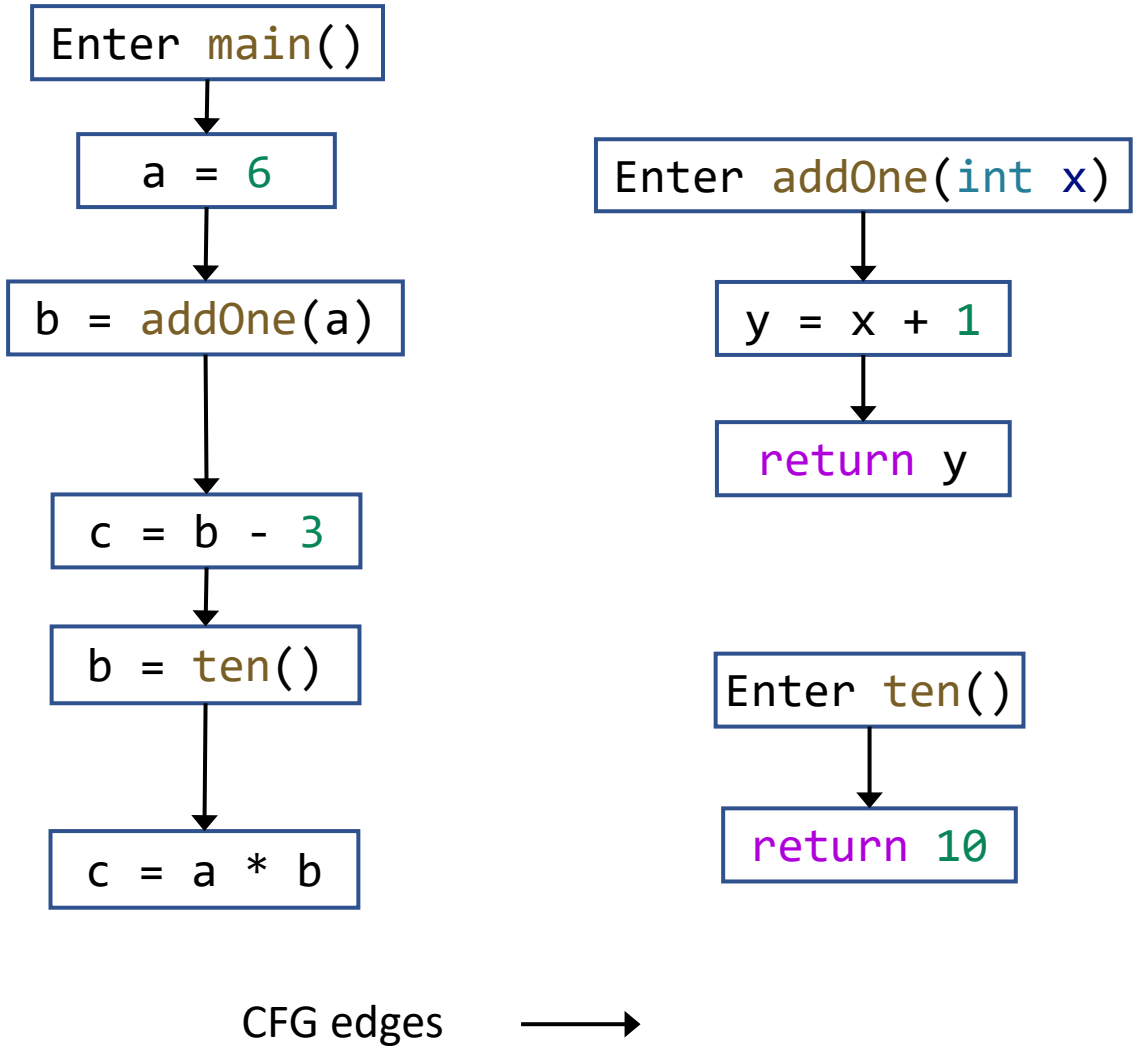

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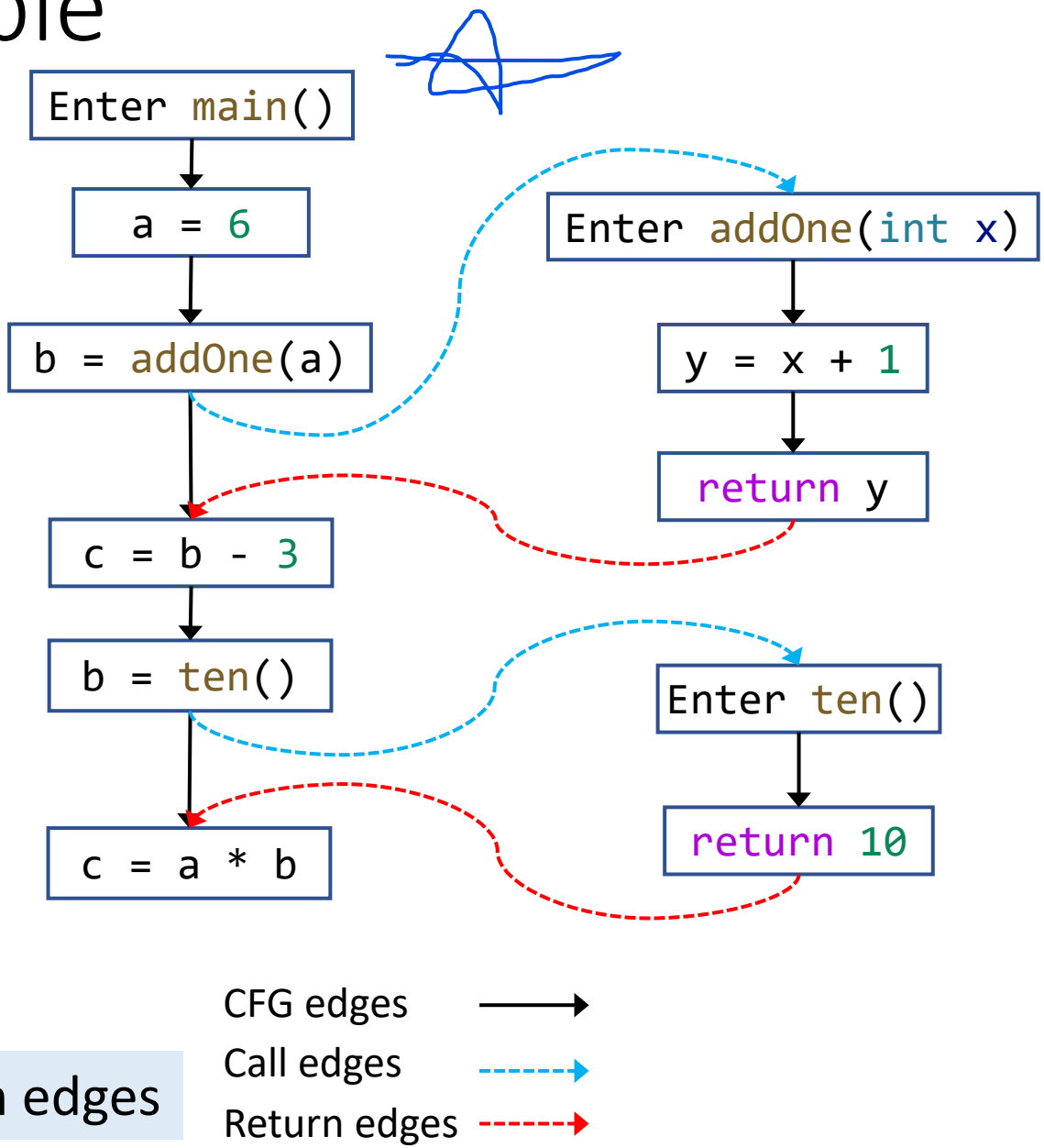

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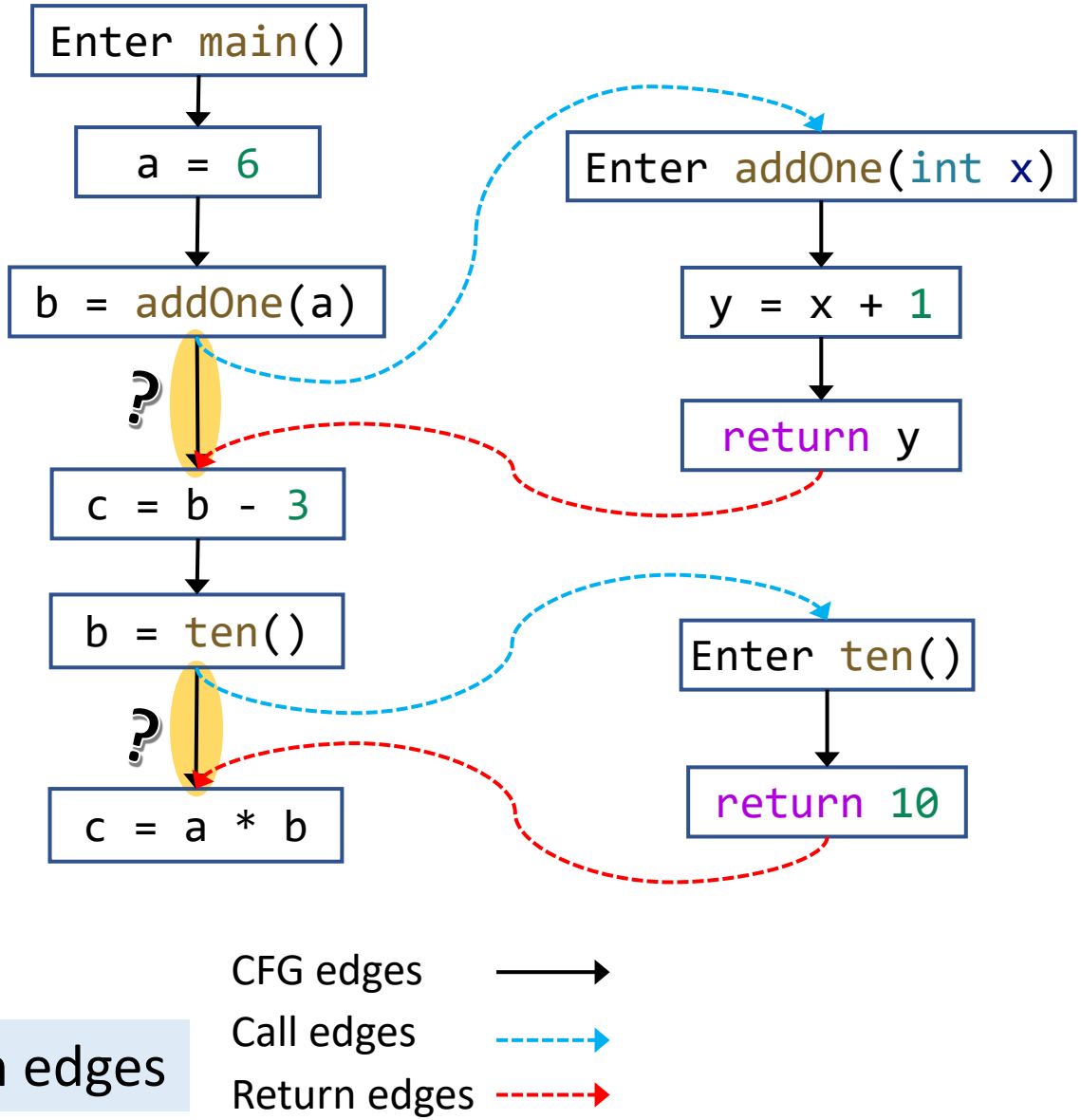

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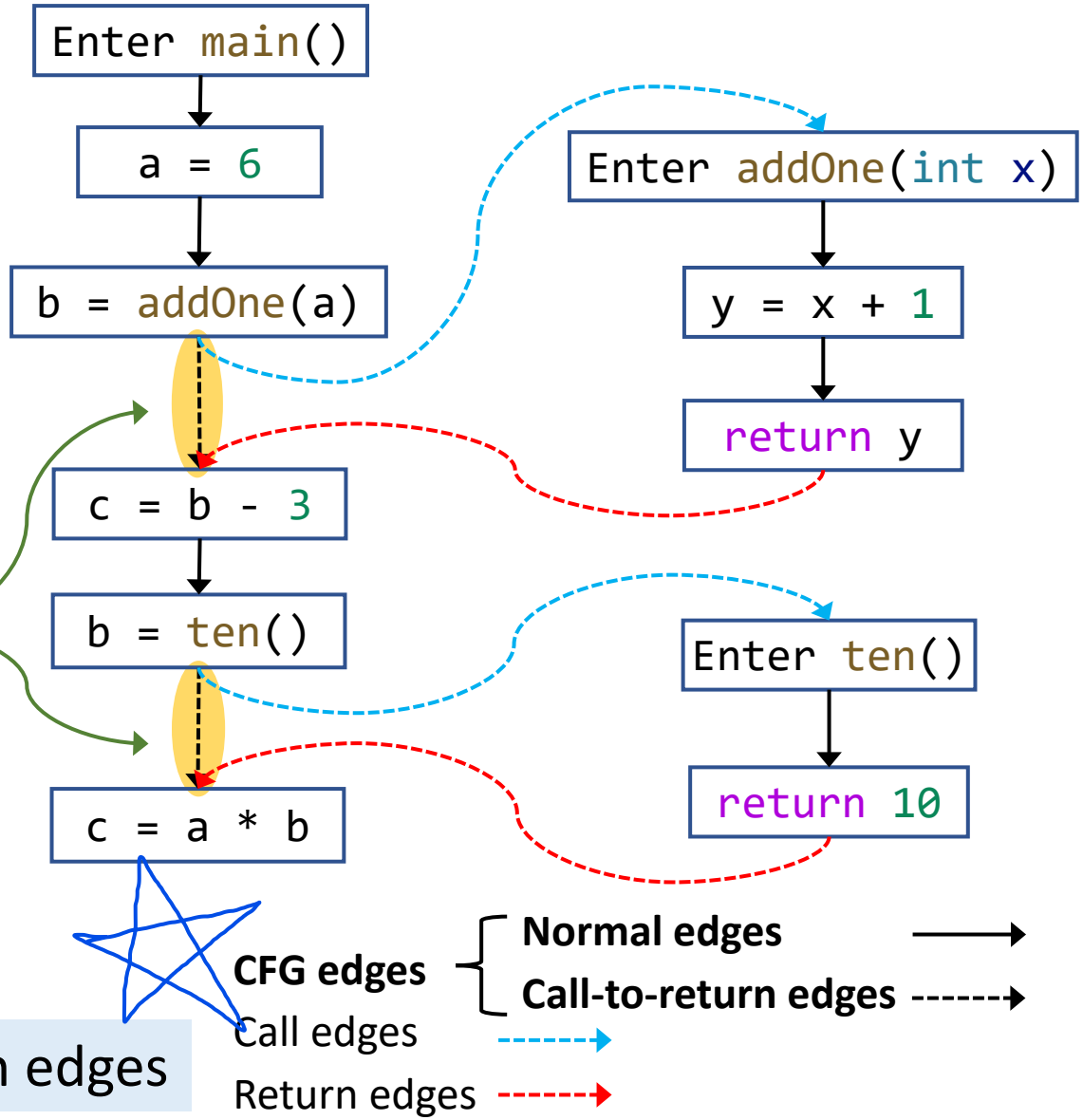

Such edges (from call site to return site) are called *call-to-return edges*

ICFG = CFGs + call & return edges

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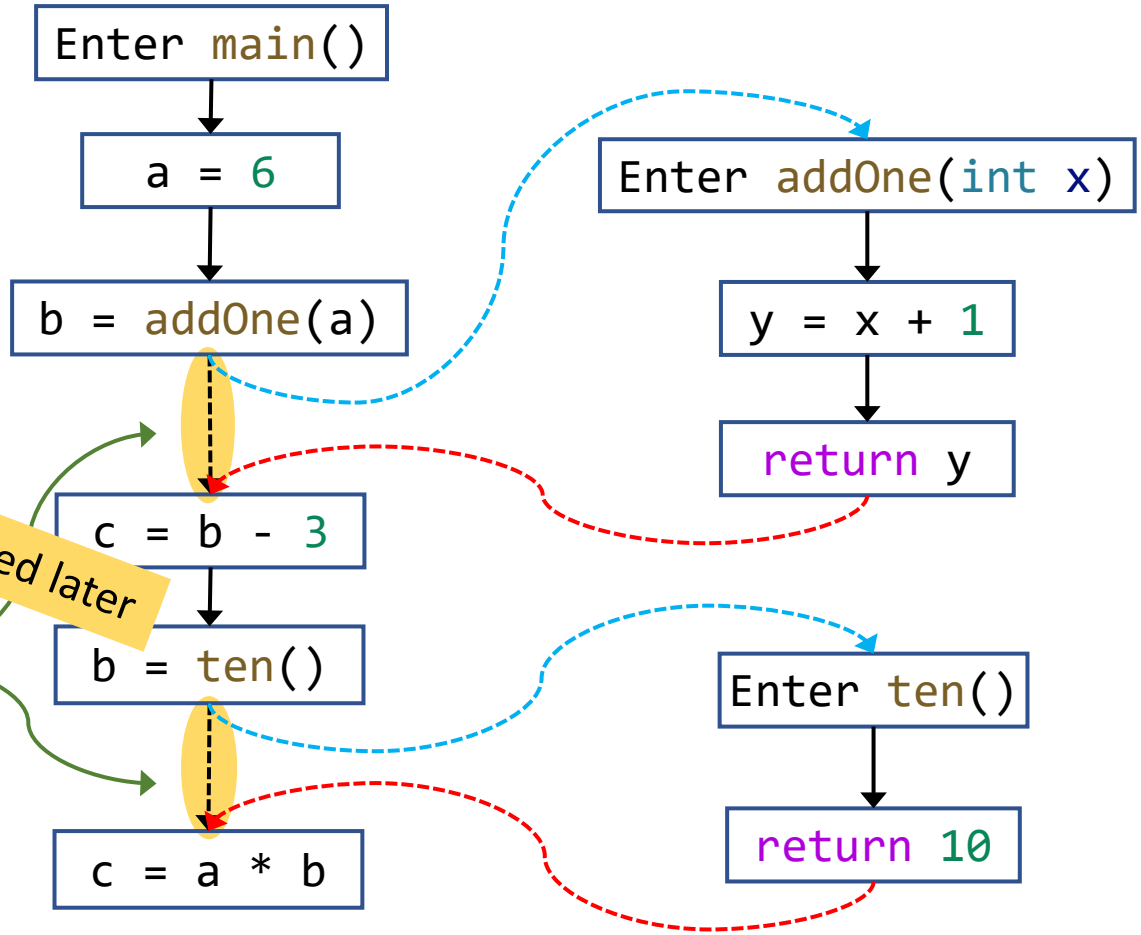

保留这种边

Will be explained later

Such edges (from call site to return site) are called *call-to-return edges*

CFG edges: **Normal edges** ⟶; **Call-to-return edges** ⟶

Call edges ⟶

Return edges ⟶

ICFG = CFGs + call & return edges

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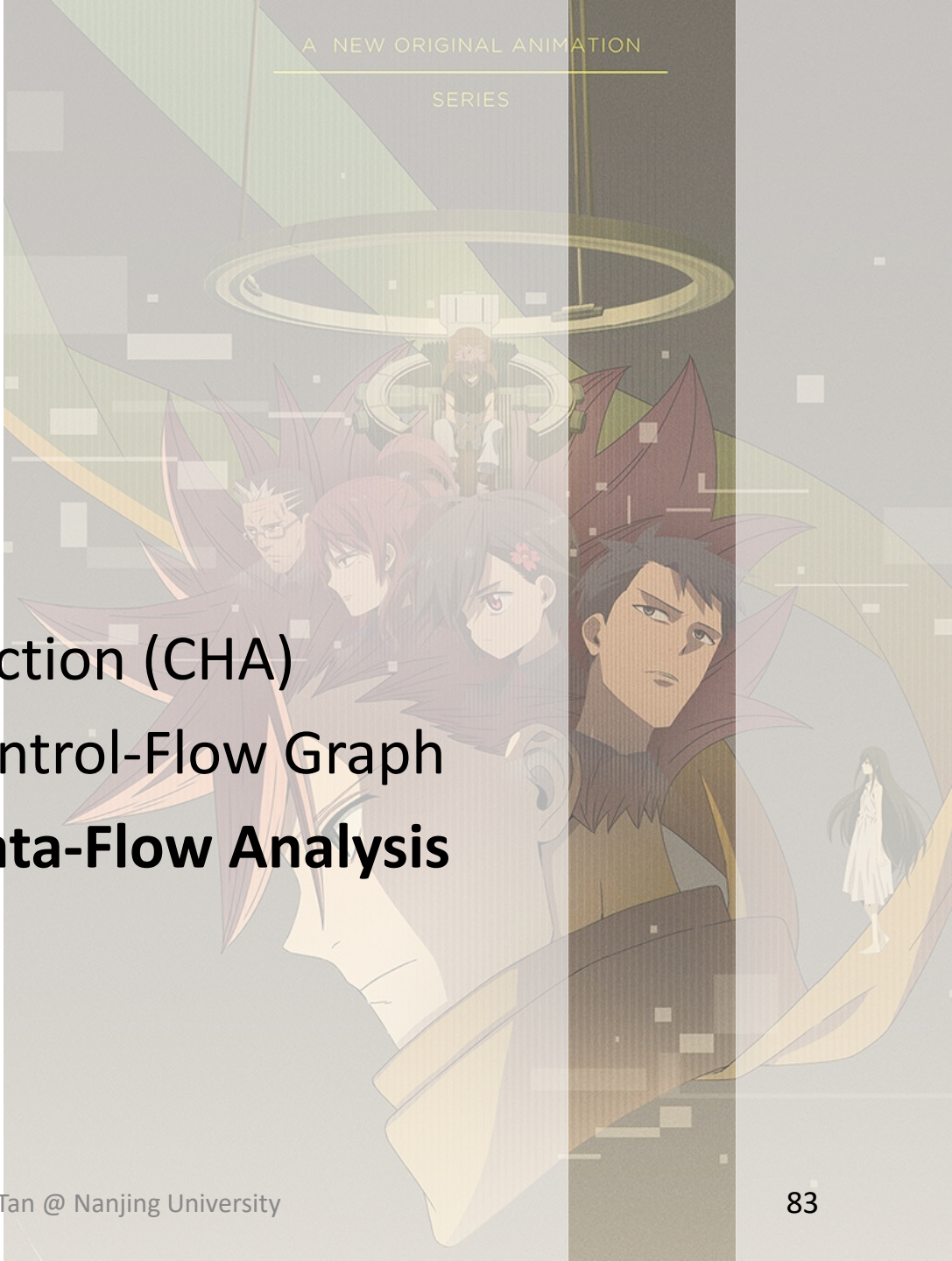

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Interprocedural Data-Flow Analysis

Analyzing the whole program with method calls based on interprocedural control-flow graph (ICFG)

	***Intra*procedural**	***Inter*procedural**
Program representation	CFG	ICFG = CFGs + call & return edges

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	*Intra*procedural	*Inter*procedural
Program representation	CFG	ICFG = CFGs + call & return edges
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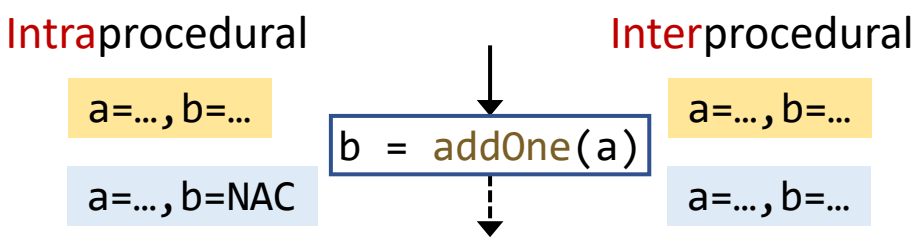
Edge transfer

- Call edge transfer: transfer data flow from call site to the entry node of callee (along call edges)
- Return edge transfer: transfer data flow from exit node of the callee to the return site (along return edges)

Interprocedural Constant Propagation

- Call edge transfer: pass argument values
- Return edge transfer: pass return values

- Node transfer: same as intraprocedural constant propagation, except that
  - For call nodes, the transfer function is *identity* function



Leave the handling of the LHS variable (return value) to edge transfer

Interprocedural Constant Propagation: An Example

```
static void main() {
  int a, b, c;
  a = 6;
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  c = b - 3;
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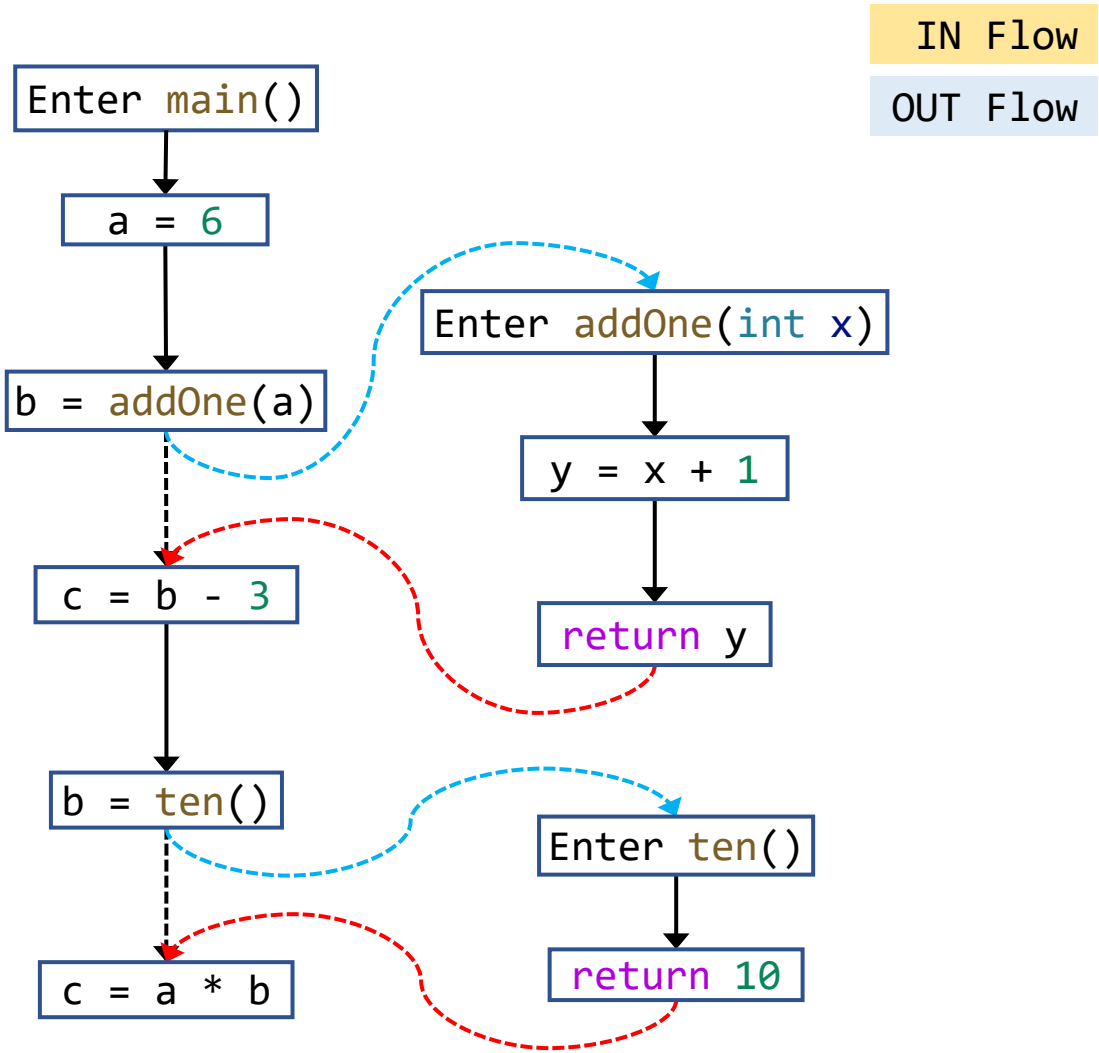

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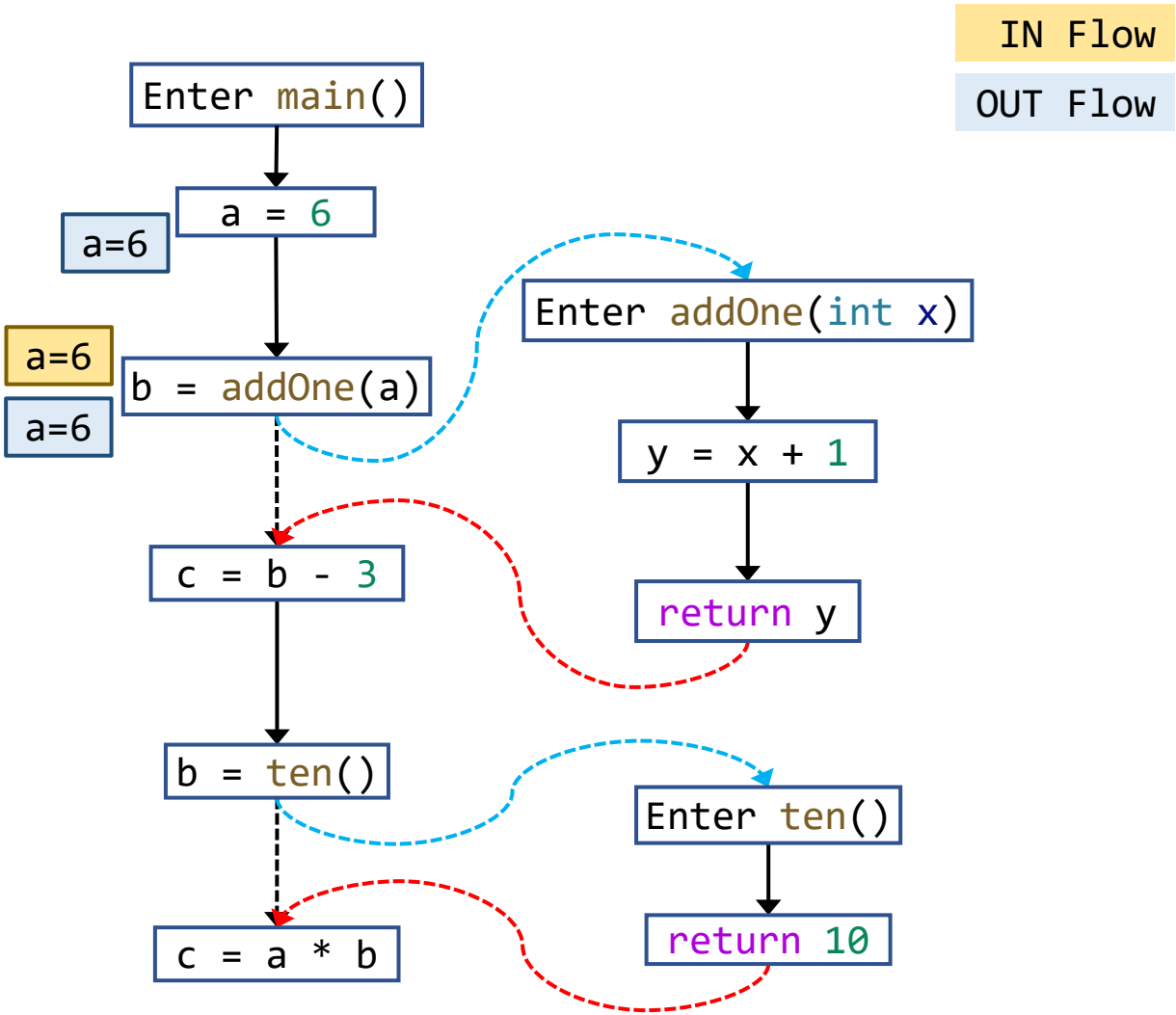

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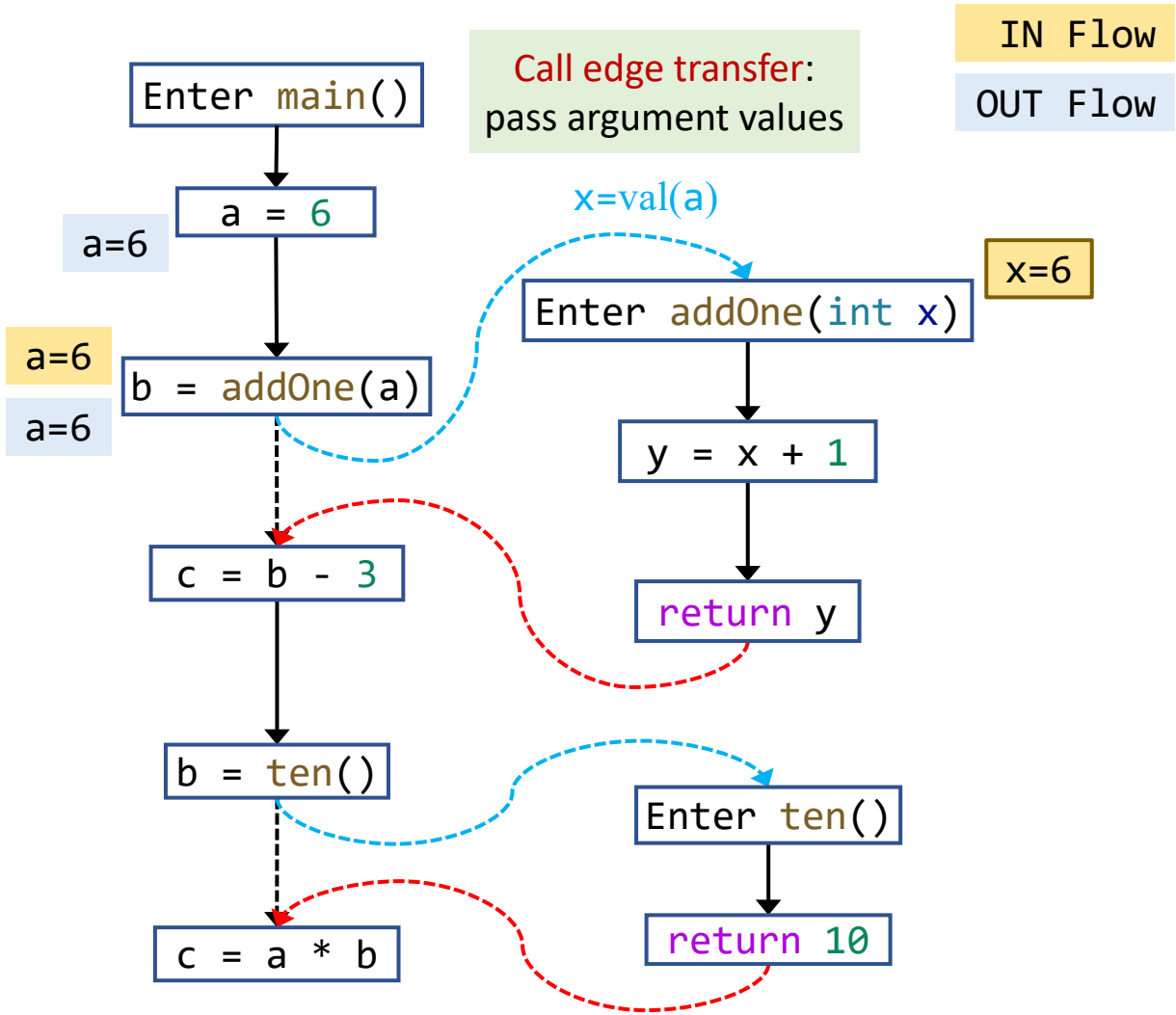

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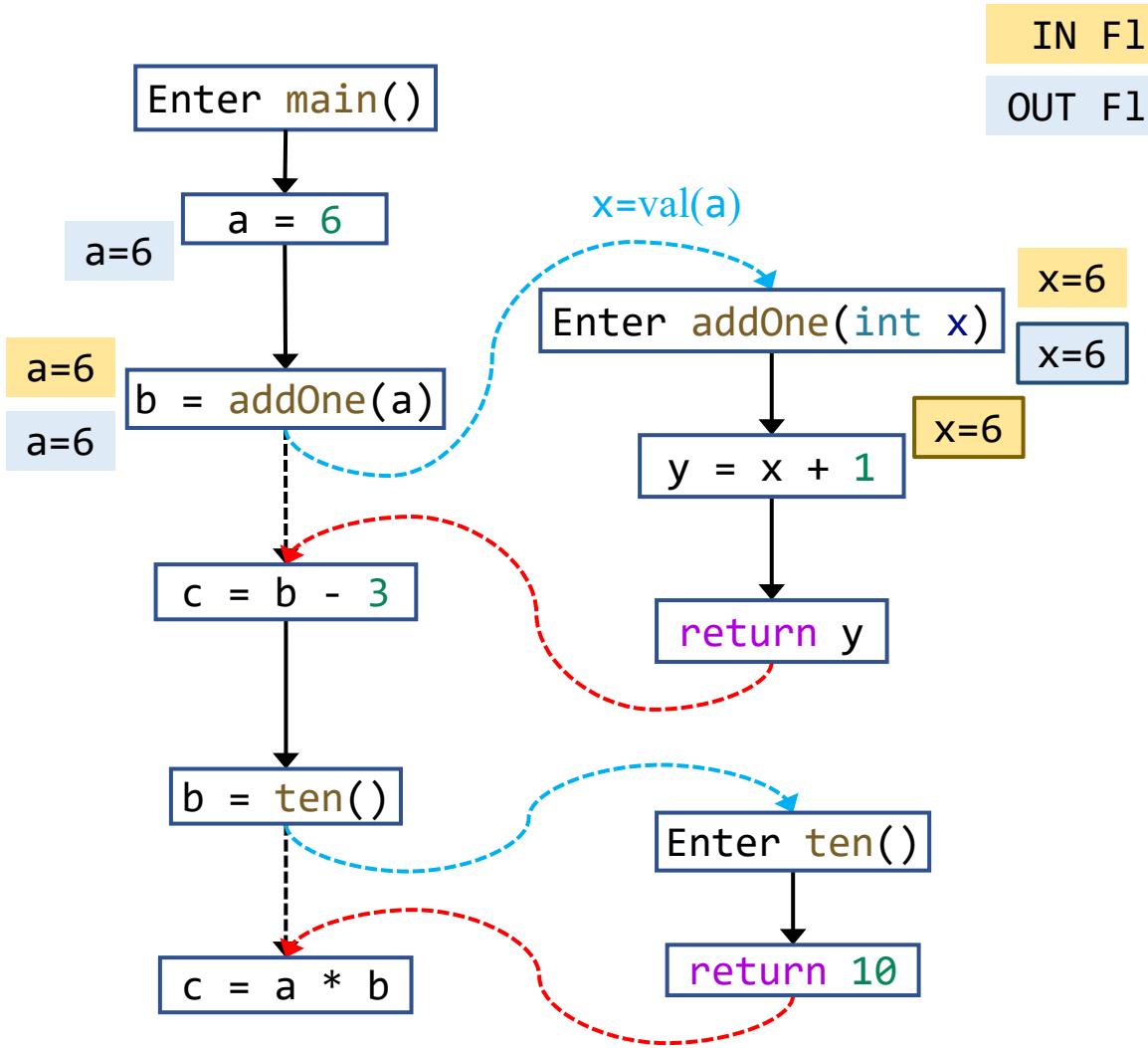

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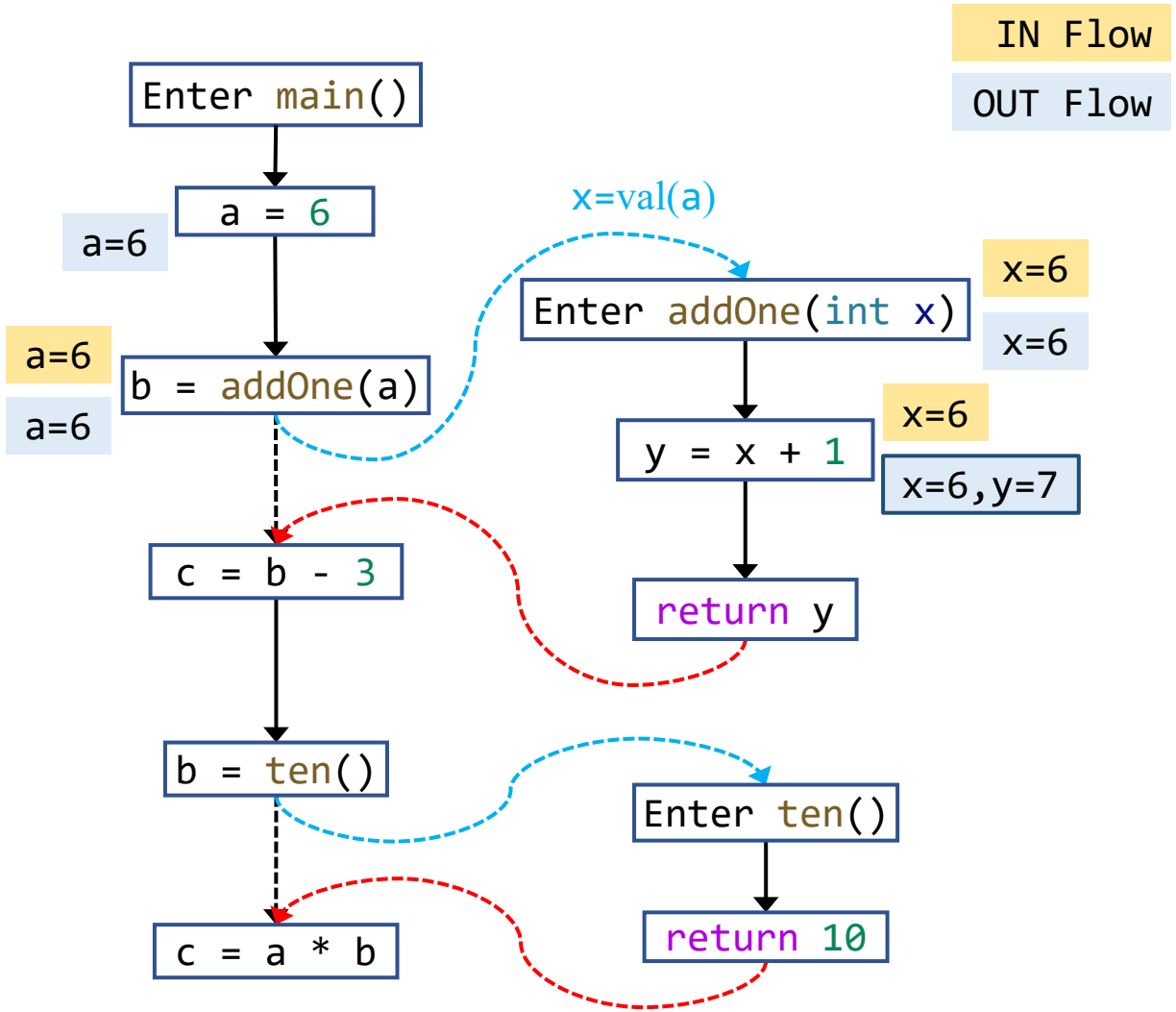

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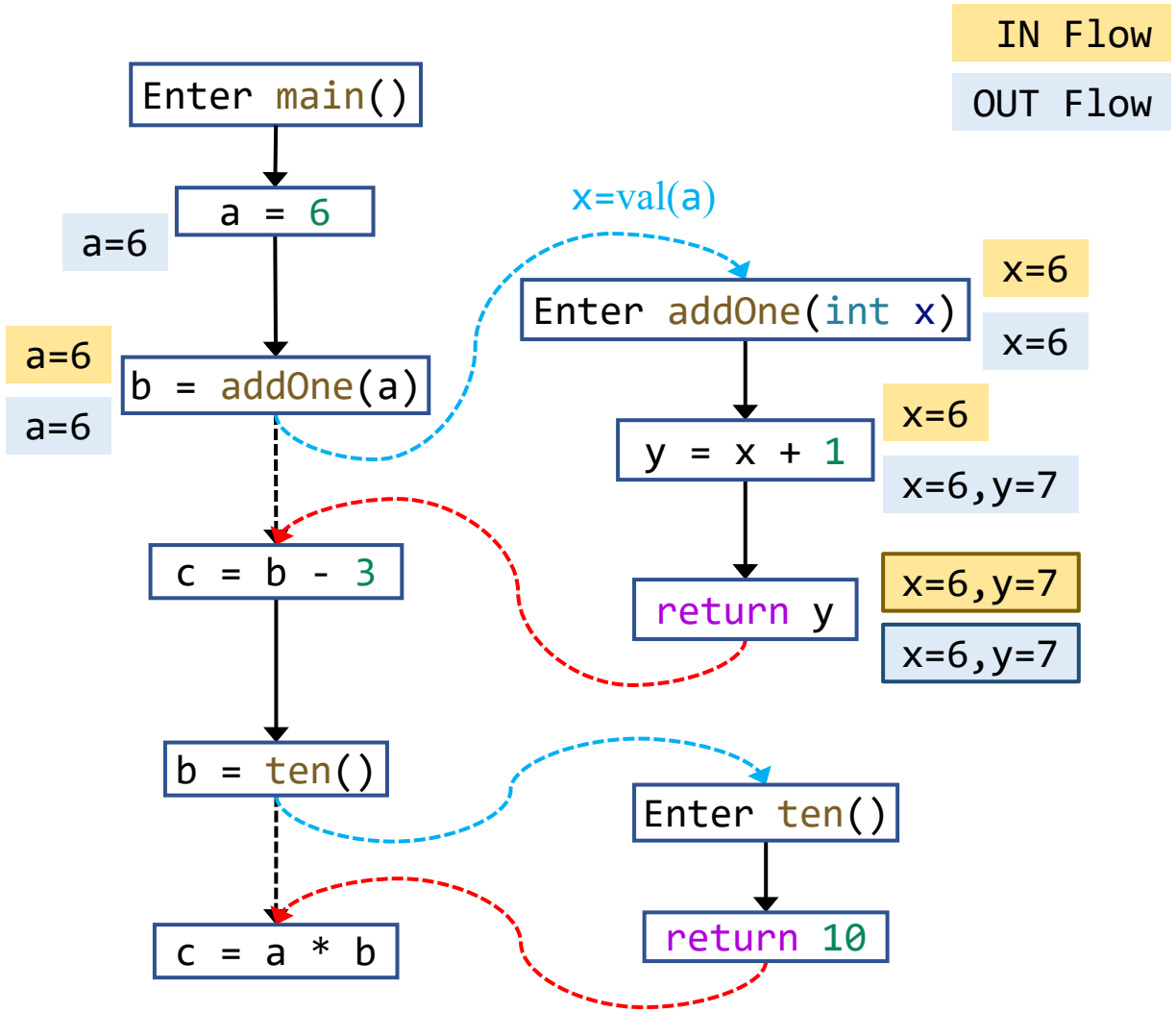

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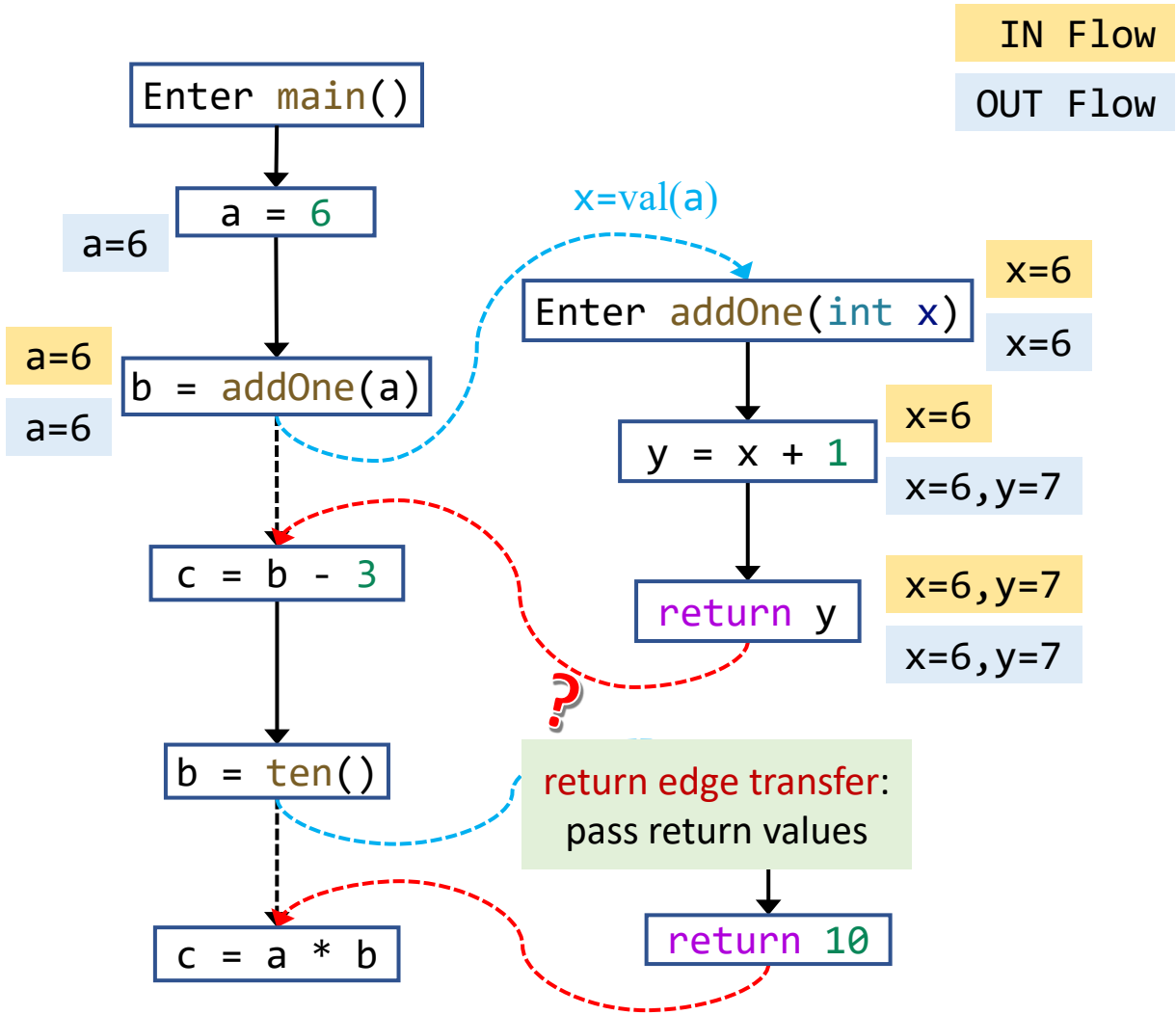

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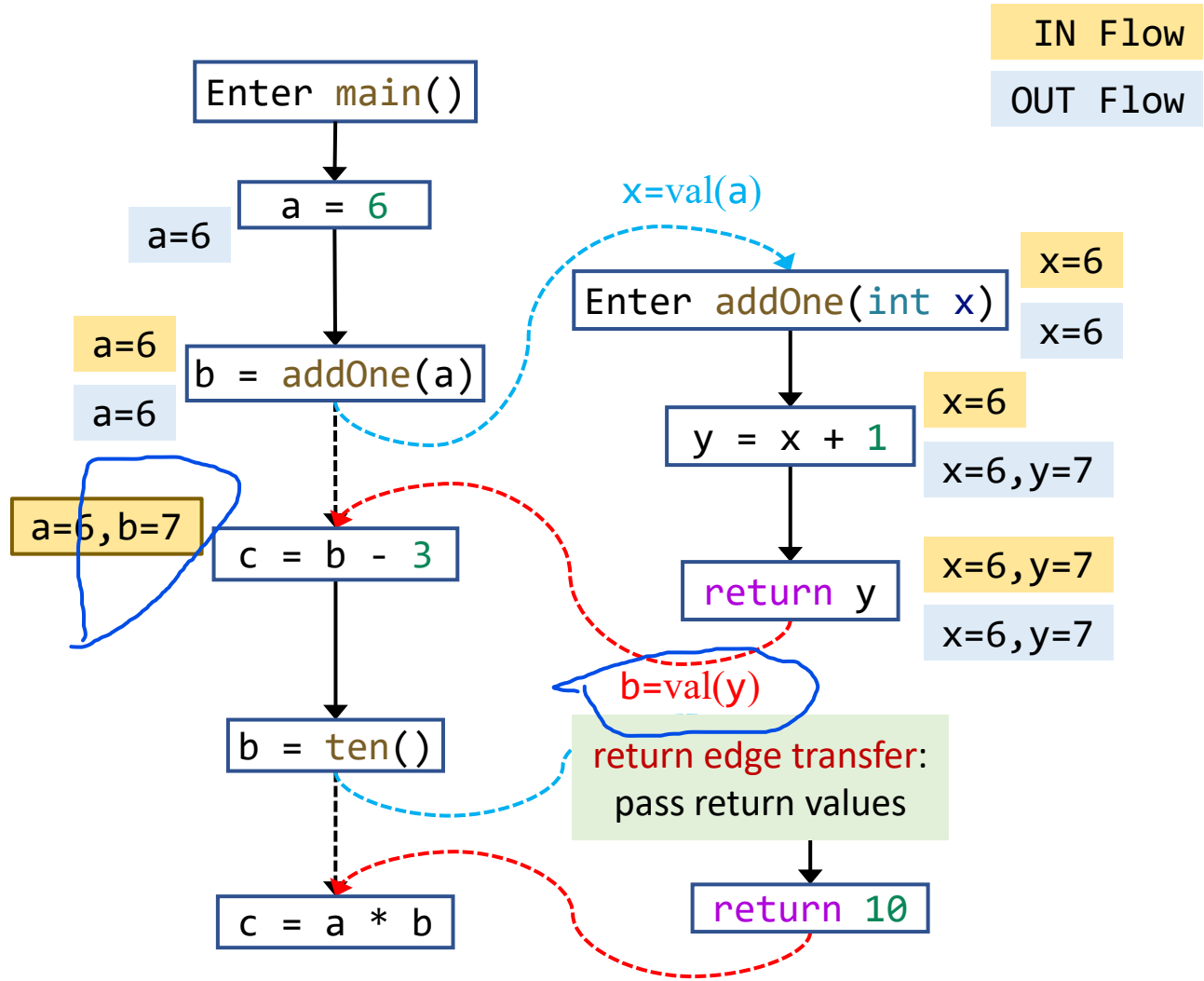

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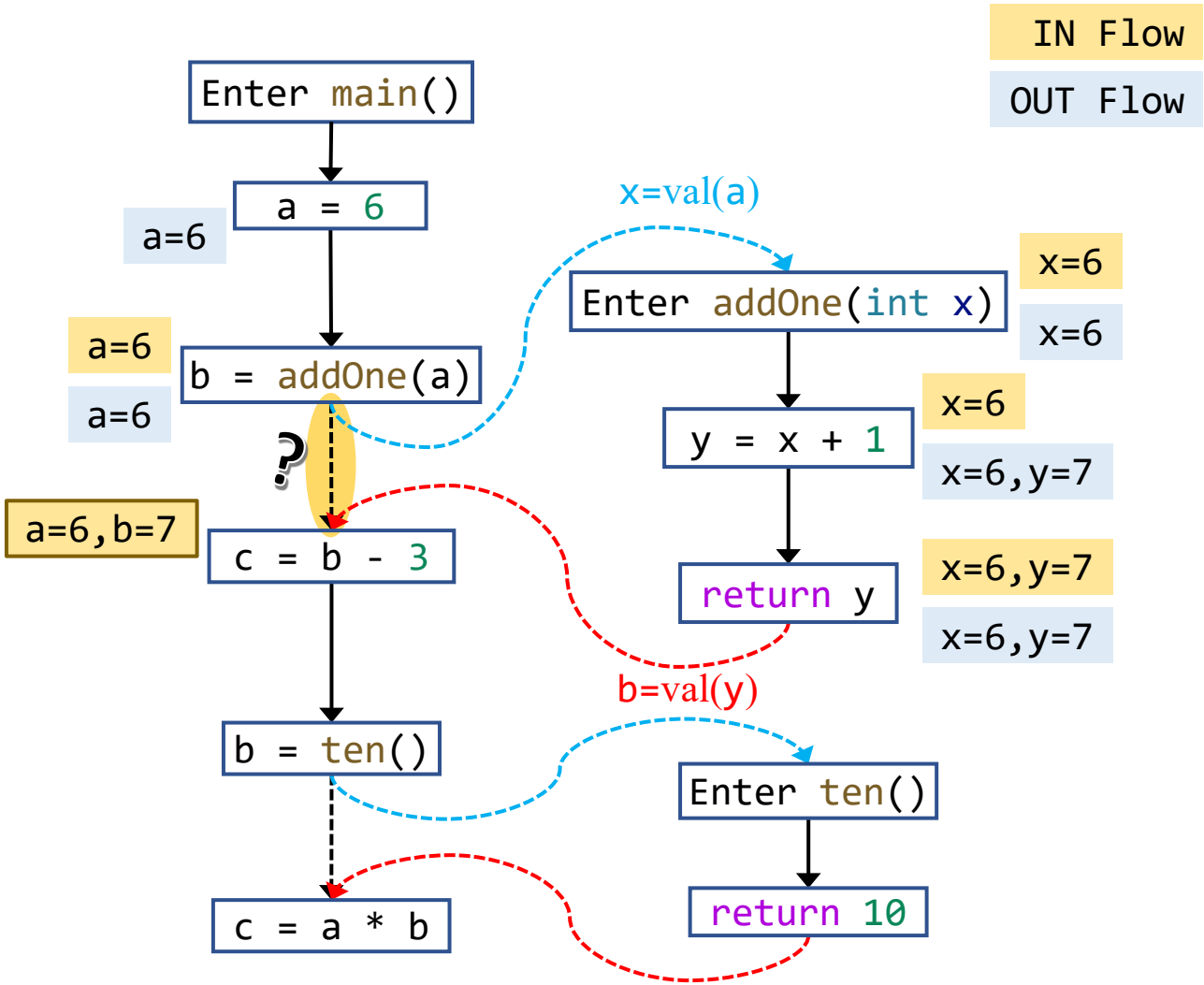

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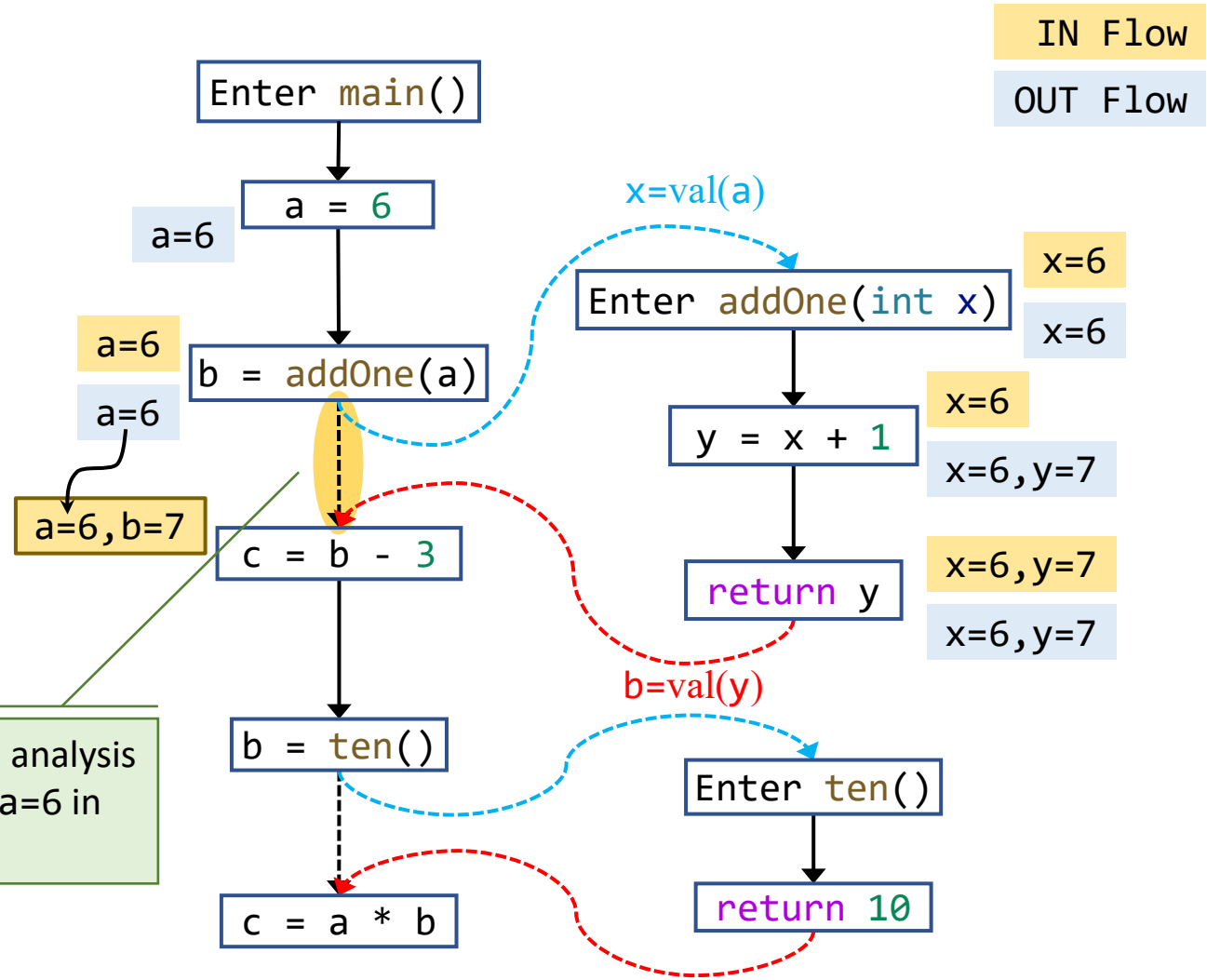

Call-to-return edges allow the analysis to propagate local data-flow (a=6 in this case) on ICFG.

可以做

函数内部的本地数据流

这里是 本地变量a

Interprocedural Constant Propagation: An Example

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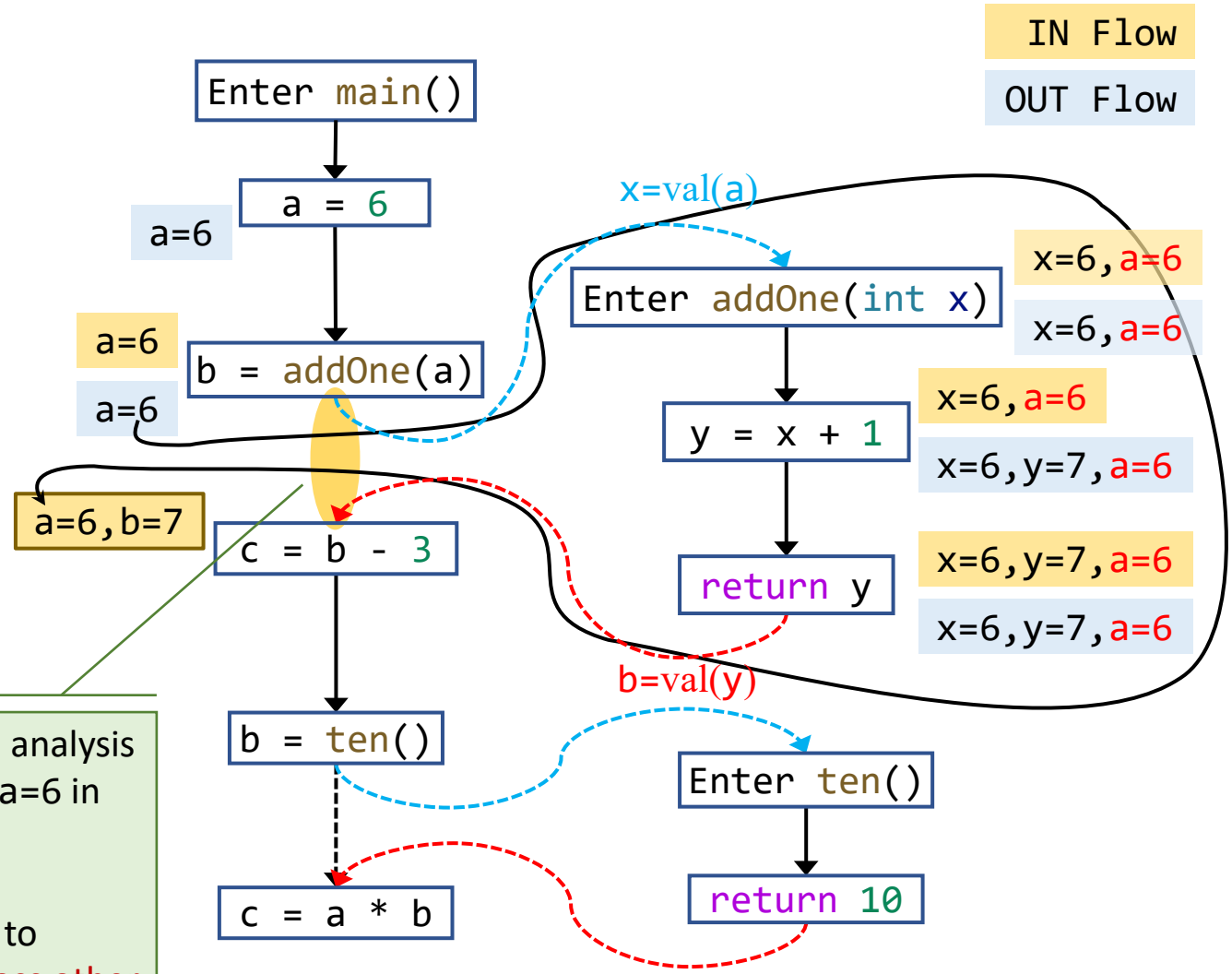

Call-to-return edges allow the analysis to propagate local data-flow (a=6 in this case) on ICFG.

Without such edges, we have to propagate local data-flow across other methods, which is very inefficient.

否则的话， 出国了

Interprocedural Constant Propagation: An Example

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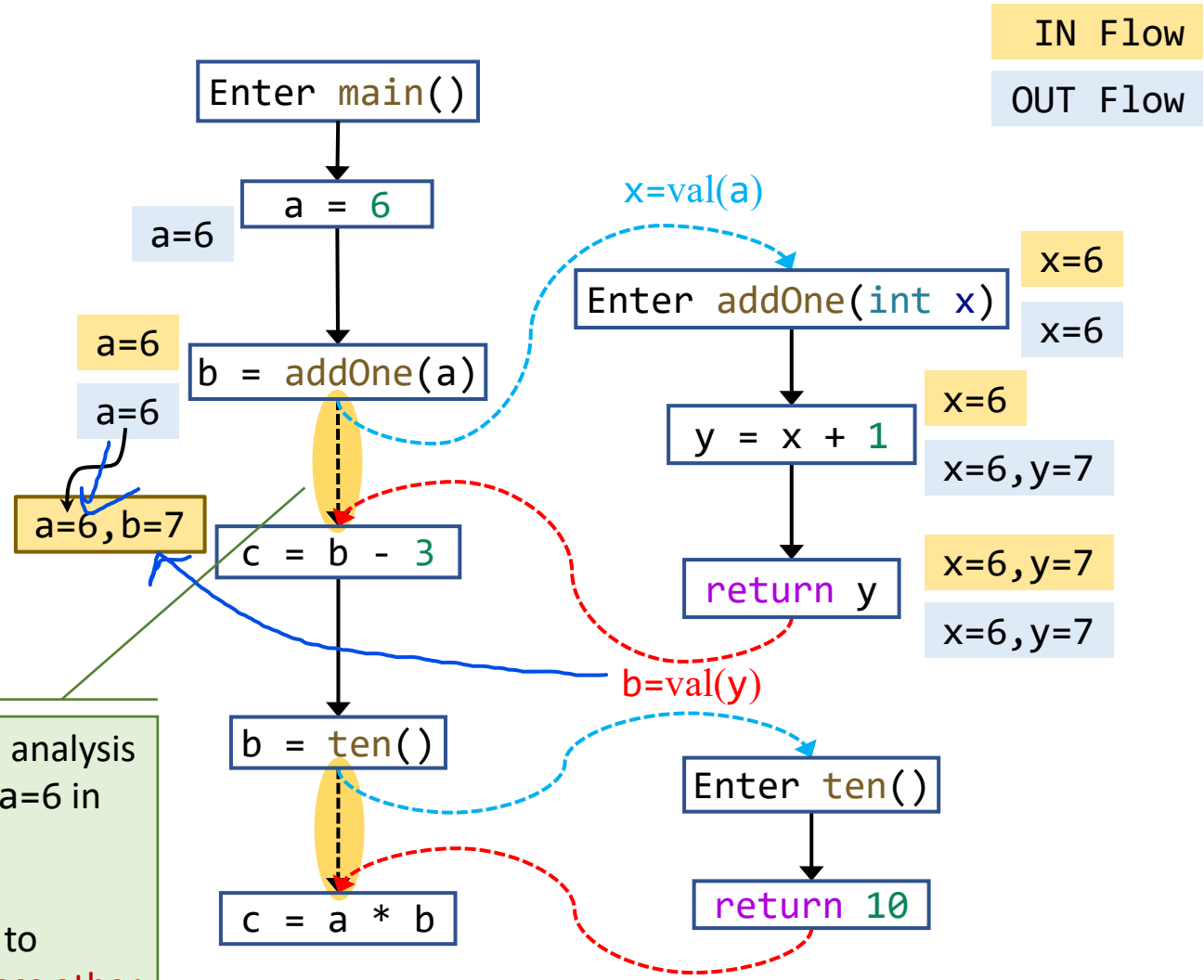

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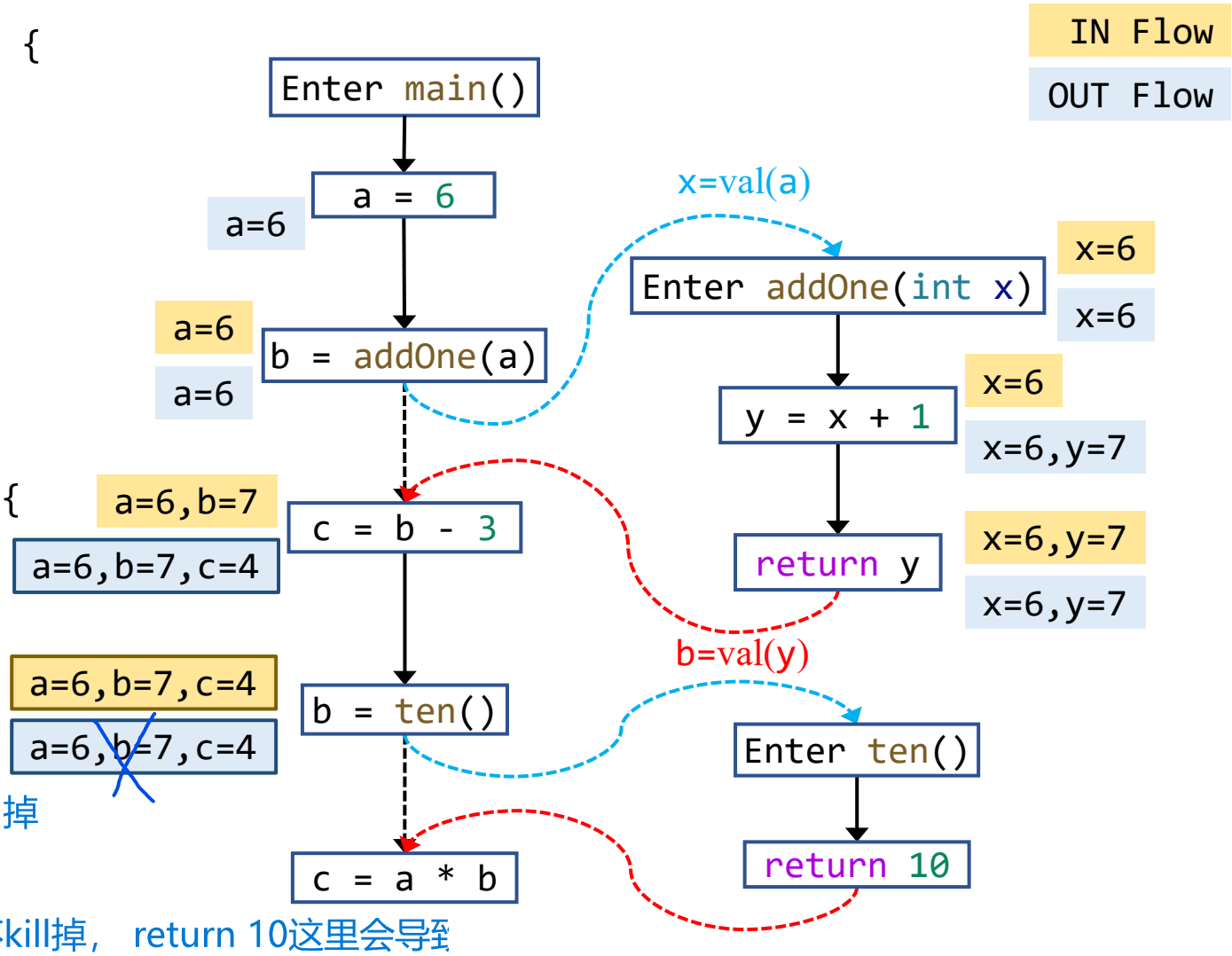
这里b要kill掉



因为如果不kill掉， return 10这里会导致

join时 b=7 和 b=10交汇
导致b=NAC

Interprocedural Constant Propagation: An Example

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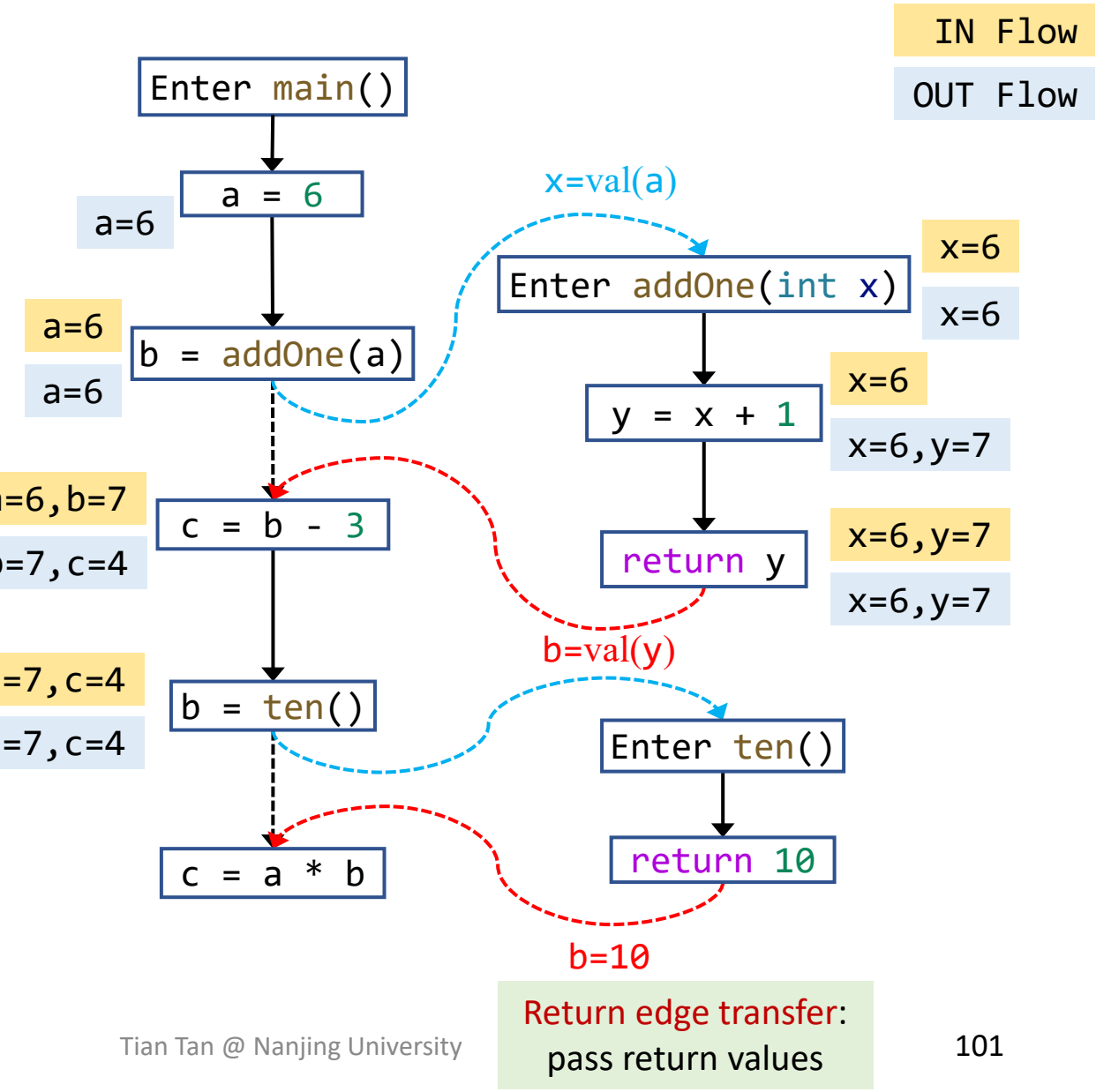

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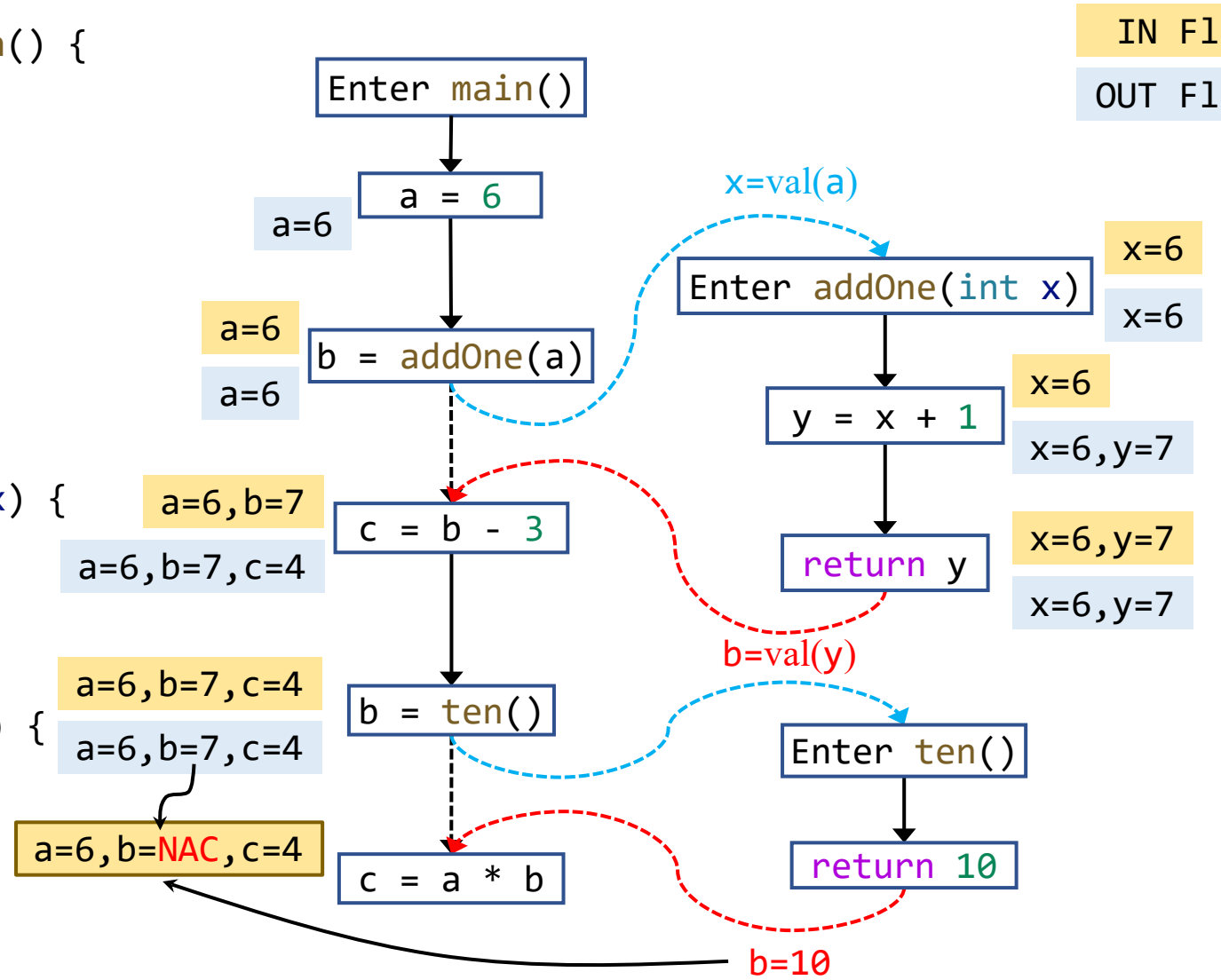

Imprecise!

Return edge transfer: pass return values

Interprocedural Constant Propagation: An Example

For call-to-return edge,
kill the value of the LHS variable of the call site. Its value will flow to return site along the return edges.
Otherwise, it may cause imprecision.

```
  b = ten();
  c = a * b;
}

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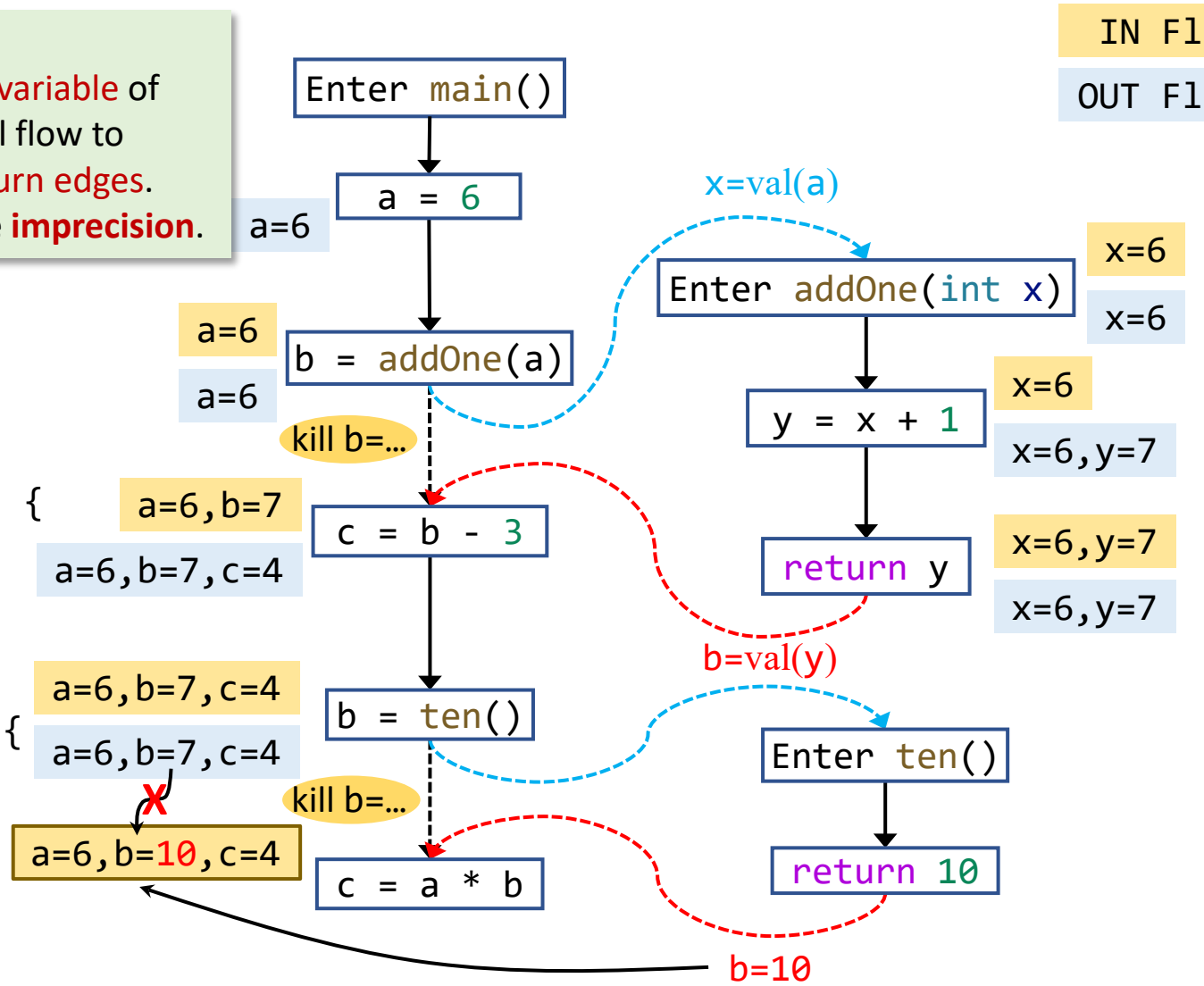
```

Precise!



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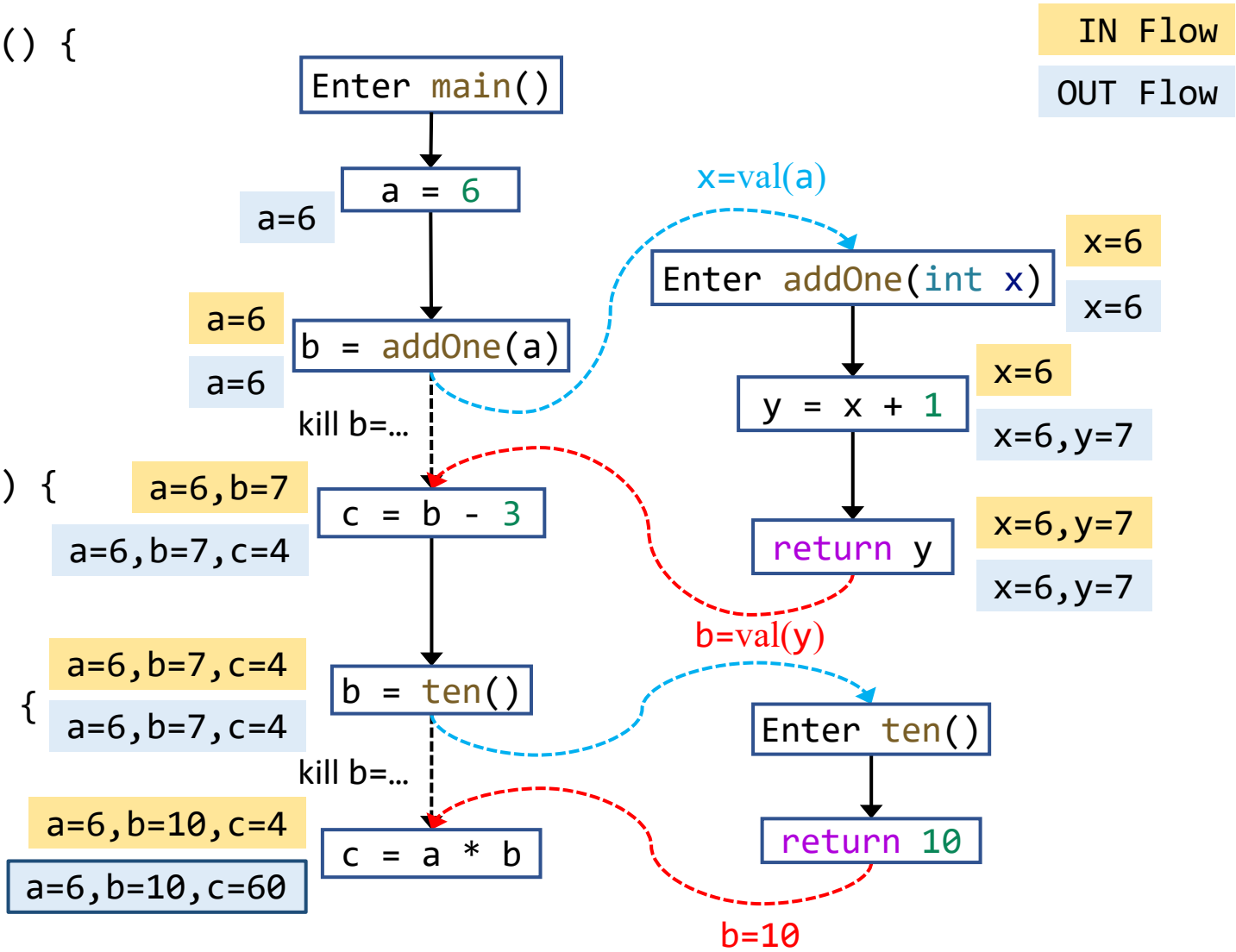

Interprocedural Constant Propagation, In Summary

- Node transfer
  - Call nodes: identity
  - Other nodes: same as intraprocedural constant propagation

- Edge transfer
  - Normal edges: identity
  - Call-to-return edges: kill the value of LHS variable of the call site, propagate values of other local variables
  - Call edges: pass argument values
  - Return edges: pass return values

过程内分析

Intraprocedural Constant Propagation: An Example

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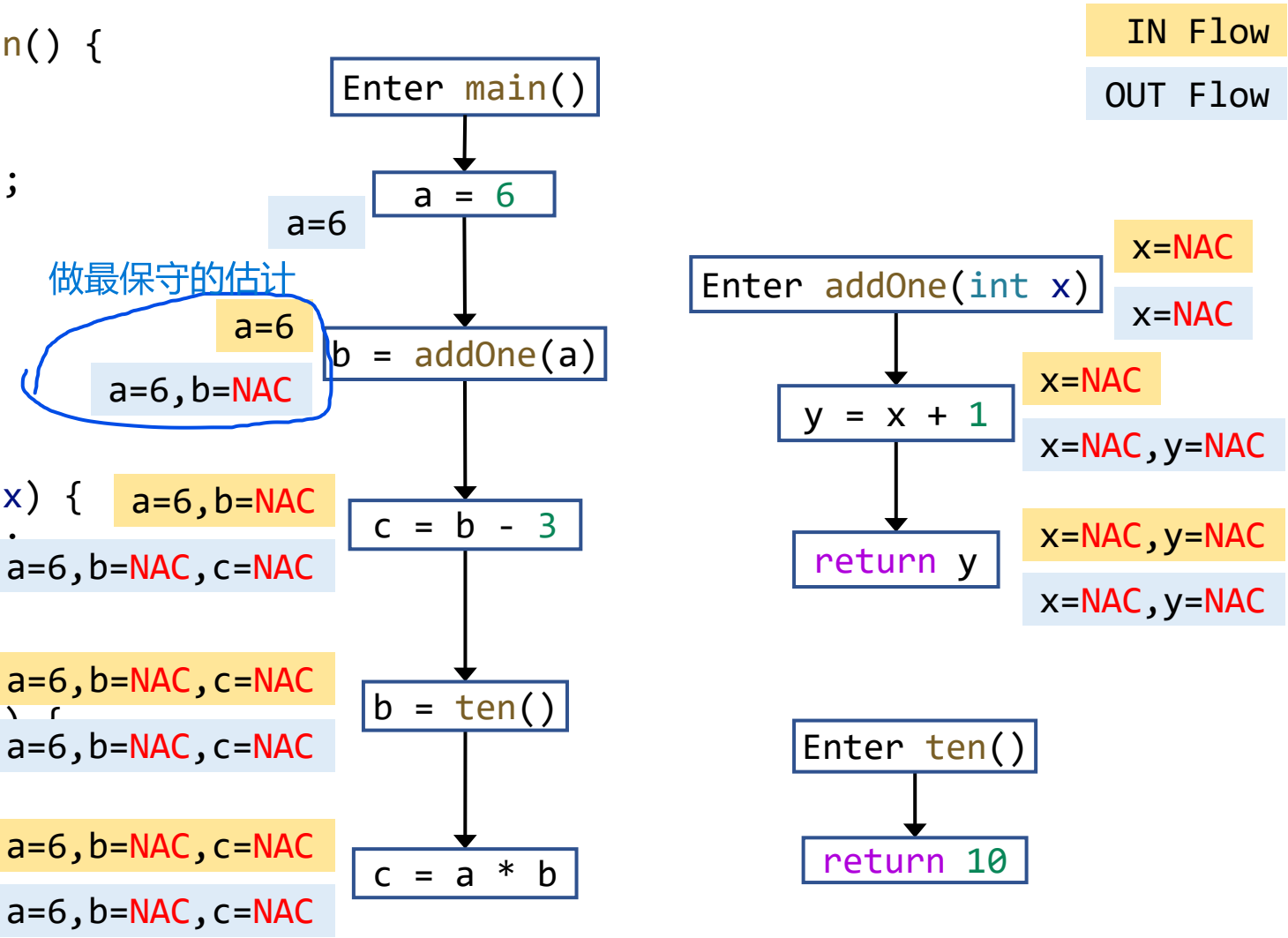

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}

static int ten() {
  return 10;
}
```

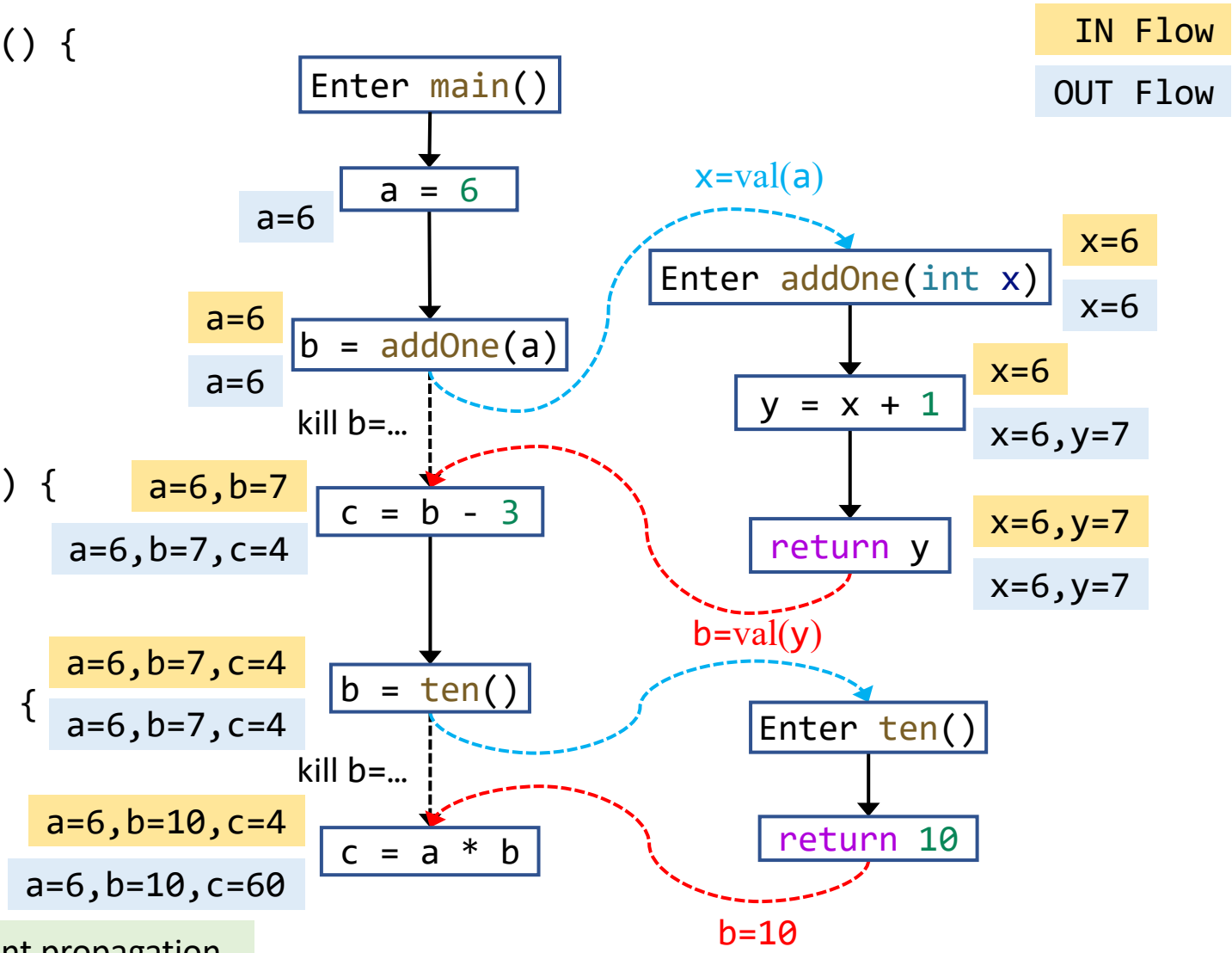

Interprocedural constant propagation is more precise than ***Intra***procedural constant propagation

The X You Need To Understand in This Lecture

- How to build call graph via class hierarchy analysis
- Concept of interprocedural control-flow graph
- Concept of interprocedural data-flow analysis
- Interprocedural constant propagation